

The Metric Blind Spot in Deep Learning-Based Brain MRI Reconstruction: A Systematic Review

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Abstract

Objective: Deep learning accelerates brain MRI four- to tenfold, but learned models can erase a lesion or invent tissue, and pixel-averaged scores such as PSNR and SSIM miss both. We review whether evaluation practice could detect this blind spot. **Methods:** Following PRISMA 2020 we searched seven databases without date limit, included 263 studies (1995–2026), appraised them with QUADAS-2 and matched instruments, and synthesised narratively. Categories came from titles, abstracts and controlled vocabulary; every prevalence figure is a floor. Duplicate screening achieved substantial agreement (Fleiss’ $\kappa = 0.877$); appraisal agreement was substantial (0.788; 0.390 where observable); extraction is unaudited. **Results:** Only 18 of 263 studies (6.8%) record a fidelity metric and a reader assessment on the same data, leaving the field’s central proxy unmeasured. Reader studies mostly measure reader-versus-reader agreement, which is weak: fastMRI 2020 concordance was 0.457 and 0.386 (Kendall’s W), improving only where SSIM had diverged. Erasing a 100 mm³ lacunar infarct moves global PSNR about 0.03 dB under the stated error model. The corpus grew fivefold; reader assessment fell from 32% to 18%, recovering to 21%. Among the four largest families the most hallucination-associated (generative, 39%) is among the least reader-evaluated (11.3%); self-supervised is higher still (47%) with no readers. One study in twenty both releases code and runs readers; none evaluated a model observer; no named dataset covers acute stroke or

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25 haemorrhage. **Conclusions:** On these floors, evaluation practice cannot certify
26 diagnostic safety. We derive five requirements safety-oriented evaluation must
27 meet.

28 *Keywords:* Deep learning, Brain MRI reconstruction, Reconstruction safety,
29 Hallucination, Image quality assessment, Systematic review

30 1. Introduction

31 Brain MRI is widely used because it can clearly distinguish different types
32 of brain tissue and reveal subtle abnormalities that are difficult to detect with
33 other imaging modalities. However, this image quality comes at the cost of long
34 acquisition times. Unlike conventional imaging methods that capture an image
35 more directly, MRI builds an image from a sequence of raw signal measure-
36 ments (samples of spatial frequencies, known as k -space) collected over time.
37 This prolonged acquisition reduces patient throughput and increases sensitivity
38 to patient motion, which can introduce artifacts and degrade image quality (Za-
39 itsev et al., 2015). To address these limitations, acceleration techniques such as
40 parallel imaging, including Sensitivity Encoding (SENSE) (Pruessmann et al.,
41 1999), and compressed sensing (Lustig et al., 2007) were developed to reduce
42 the amount of data that must be collected. More recently, deep learning has
43 enabled further acceleration by reconstructing high-quality images from incom-
44 plete MRI measurements, through cascaded convolutional networks (Schlemper
45 et al., 2017), model-based unrolled networks (Hammernik et al., 2018; Aggar-
46 wal et al., 2018), generative adversarial networks (Yang et al., 2018), and score-
47 based diffusion models (Song et al., 2021), at accelerations from fourfold to, in
48 exploratory work, an order of magnitude (Radmanesh et al., 2022).

49 The reconstructed images look convincing, and that is the problem. Small
50 changes in the input can produce large errors in the output (Antun et al., 2020;
51 Gottschling et al., 2025). Four failure types recur across the studies we re-
52 viewed: strong priors smooth away faint structure; motion causes ghosting the
53 model never saw in training (Sommer et al., 2020); a change of scanner or sam-

54 pling degrades results in ways that in-distribution testing misses (Güngör et al.,
 55 2023); and generative models can invent plausible tissue (Zhou et al., 2024).
 56 The scores used to certify these methods, peak signal-to-noise ratio (PSNR)
 57 and structural similarity (SSIM) (Wang et al., 2004), agree only partially with
 58 radiologist quality judgement even on conventional image degradations (Mason
 59 et al., 2020); and SNR, though reproducible across observers, tracks no-reference
 60 quality models only moderately (Yu et al., 2018), so an image can score well
 61 while the error that matters goes unrecorded. We call this the **metric blind**
 62 **spot**: standard image-quality metrics cannot catch these small but clinically se-
 63 rious errors, because a lesion of a few cubic millimetres barely moves an average
 64 taken over millions of voxels.

65 Parts of the field have responded, training quality metrics on radiologist
 66 rankings (Tang et al., 2025), attaching uncertainty maps to reconstructions
 67 (Khawaled and Freiman, 2024), or running reader studies that test an acceler-
 68 ated protocol against a reference one directly (Lang et al., 2024; Radmanesh
 69 et al., 2022). Causal and counterfactual methods can ask the most direct ques-
 70 tion, whether a lesion survives a controlled change of acquisition, but have
 71 mostly been applied to classification rather than reconstruction (Pawlowski
 72 et al., 2020; Castro et al., 2020). These responses share a root cause. Ac-
 73 celeration works because a learned prior fills in information the measurement
 74 does not contain, and that same filled-in information can hide a real finding or
 75 create a false one, while the field certifies the result with averaged pixel scores
 76 that cannot see either event.

77 To our knowledge no systematic review has assembled the evidence on this
 78 problem. Existing surveys cover architectures and benchmark performance
 79 (Liang et al., 2020; Montalt-Tordera et al., 2021); none asks how often con-
 80 ventional metrics miss clinically important errors, why they miss them, or what
 81 evaluation would have to look like to catch them. We address that through
 82 three questions:

83 **RQ1** *How reliably do conventional fidelity metrics track radiologist diagnostic*

84 *ratings in deep learning-based accelerated brain MRI, and how wide is the*
85 *evaluation reporting gap in the literature?*

86 **RQ2** *What underlying mathematical mechanisms cause standard metrics to*
87 *overlook localized pathology erasure and generative hallucinations in deep*
88 *learning-based reconstruction?*

89 **RQ3** *What can each evaluation paradigm in current use establish about prior-*
90 *induced diagnostic error in deep learning-based MRI reconstruction, and*
91 *what requirements for safety-oriented evaluation follow from those limits?*

92 We contribute: (C1) a synthesis of what 263 primary studies report about
93 agreement between pixel-wise metrics and radiologist judgement; (C2) four
94 mathematical mechanisms that explain the divergence, with an exact identity
95 bounding the PSNR cost of erasing a lesion; (C3) an analysis of *where* the
96 evaluation gap comes from, showing that it co-occurs with a division of evalu-
97 ation practice between research communities and with a benchmark infrastruc-
98 ture lacking acute pathology; and (C4) five requirements that safety-oriented
99 evaluation must meet. The auditing framework these requirements motivate is
100 developed and evaluated in a companion methods paper, not here.

101 Three boundaries should be stated at the outset, because they determine
102 what the findings can and cannot be asked to support. First, this is a review of
103 *evaluation practice*, not of reconstruction performance: we count what studies
104 measured, not how well their methods performed, and nothing here estimates
105 how often a deployed reconstruction actually erases a lesion. Second, the mech-
106 anisms in RQ2 are mathematical properties of the metrics, derived under a
107 stated error model; they establish that a focal error *can* pass unrecorded, not
108 that it commonly does. Third, prevalence counts rest on labels extracted from
109 what studies report, so every one of them is a lower bound on practice rather
110 than an estimate of it, and we treat the direction of that bias explicitly in Sec-
111 tion 5.2. The contribution is a diagnosis of what the field’s current instruments
112 are capable of detecting, which is logically prior to, and considerably cheaper
113 to establish than, a measurement of how often they fail.

114 2. Related Work

115 Four strands bear on this review, and the claim that none of them has
116 assembled the evidence we assemble here is a claim about each in turn.

117 *Surveys of reconstruction methods.* Reviews of deep learning MRI recon-
118 struction catalogue architectures, acceleration factors and benchmark accuracy
119 (Liang et al., 2020; Montalt-Tordera et al., 2021; Knoll et al., 2020a; Kazer-
120 ouni et al., 2023). They are organised by method, and their outcome variable
121 is reconstruction fidelity against a fully sampled reference. Safety and failure
122 modes appear as caveats in discussion sections rather than as an object of syn-
123 thesis, and none of them tabulates what evaluation each primary study actually
124 performed, which is the quantity this review extracts.

125 *Task-based image quality assessment.* The tradition that ought to have sup-
126 plied the answer established, decades before deep learning, that image quality is
127 meaningful only relative to a diagnostic task. Barrett and colleagues formalised
128 the observer as part of the measurement: an image is good if an observer per-
129 forming a specified task on it achieves a high figure of merit, typically the
130 area under an ROC curve for a detection task, and the mathematical (ideal or
131 channelised Hotelling) observer supplies a computable surrogate for the human
132 when the task and image statistics are specified (Barrett et al., 1993; Barrett
133 and Myers, 2004). ICRU Report 54 made the same argument the international
134 reference framework for assessing medical imaging quality (International Com-
135 mission on Radiation Units and Measurements, 1996). That framework already
136 contains the concept the reconstruction literature lacks: a detectability index
137 for a specified lesion, which is by construction sensitive to exactly the focal
138 change a whole-volume L_2 score averages away. It has been carried into the
139 dose- and protocol-optimisation literatures of other imaging modalities, but it
140 has not crossed into deep learning MRI reconstruction: no study in this cor-
141 pus evaluated a model observer, and our first requirement (R1, Section 4.7) is
142 in substance a restatement of the task-based position for a field that has not
143 adopted it. The reason it has not is worth naming: model observers require a

specified signal and a statistical description of the background, and the null-space structure that makes learned reconstruction dangerous is precisely what makes those two ingredients hard to specify.

Failure-mode-specific work. Artifact detection, motion correction and robustness under distribution shift are each addressed in depth, but in isolation, by literatures that do not share an evaluation vocabulary (Esteban et al., 2017; Sommer et al., 2020; Güngör et al., 2023). Analyses of hallucination in tomographic reconstruction establish that prior-induced structure lives in the null space of the forward operator, so it is invisible to any measurement-space consistency check (Bhadra et al., 2021; Gottschling et al., 2025), and causal and counterfactual methods for medical imaging supply the machinery to intervene on image content and ask whether a finding survives (Pawlowski et al., 2020; Castro et al., 2020; Sanchez et al., 2022; Peng et al., 2026). Neither line has been connected to review-level evidence on how reconstruction is evaluated in practice.

What is new in the four mechanisms of RQ2, and what is not. Because Section 4 states four mechanisms formally, their provenance should be set out plainly rather than left for a reader to reconstruct. Two are elementary properties of the objective rather than findings: spatial error averaging follows from the definition of a mean over the image volume, and loss symmetry from the fact that squaring discards the sign of the residual. Two are established results we restate: reference dependence is standard in the image-quality literature, and the null-space account underlying the realism–reward trade-off is due to Bhadra et al. (2021) and Gottschling et al. (2025). What this review adds is not the mechanisms but their quantification and their assembly: the identity at Equation (2), which converts any lesion volume into the PSNR movement its erasure would cause, and the setting of all four against what a corpus of 263 studies is recorded as measuring. We claim the assembly and the bound, not the underlying results.

Benchmark and annotation infrastructure. The raw k -space resource the field benchmarks on is fastMRI (Knoll et al., 2020b), whose brain release fol-

lowed the knee data described in that report, and an annotation layer is emerging on top of it: fastMRI+ adds expert pathology labels to the fastMRI brain and knee data (Zhao et al., 2022), and curated stroke-lesion datasets exist for segmentation (ISLES (Hernandez Petzsche et al., 2022), ATLAS (Liew et al., 2022)). None of these supplies raw k -space paired with acute vascular pathology, which is the combination reconstruction evaluation needs, and Section 4.5 shows that this is not an incidental gap but the binding constraint on the whole evaluation problem.

What no strand supplies is the join: a systematic account of how often the field measures metric–reader agreement, why the metrics fail when it does not, and what each evaluation paradigm can therefore establish. The four strands are not merely separate; they are held apart by what each can access. The survey and hallucination-analysis literatures work on datasets with raw k -space and no clinically verified pathology; the task-based tradition needs a specified signal and a labelled task, which those datasets do not carry; and the clinical reader literature has the pathology and the readers but no access to the raw data on which the mechanisms operate. Section 4.5 shows this is not an abstraction but a measurable division in the corpus itself. Supplementary Table S4 sets out each secondary study we identified against the questions asked here.

3. Methodology

This systematic review was conducted in accordance with the PRISMA 2020 statement (Page et al., 2021); the completed checklist, search vocabulary, extraction schema, and analysis scripts are available in the supplementary material and on the Open Science Framework (<https://doi.org/10.17605/OSF.IO/832Z6>; timestamped deposit <https://doi.org/10.17605/OSF.IO/EUXA8>).

Protocol and registration status. The protocol, covering the research questions, eligibility criteria, search strategy, screening procedure, classification rubric and synthesis approach, was fixed in the project repository before screening began. In the terms of PRISMA 2020 item 24a, **this review was**

204 **not prospectively registered and carries no registration number:**
 205 PROSPERO’s scope excludes reviews without direct clinical health outcomes,
 206 and no equivalent prospective registry covers methodological reviews of this
 207 kind. The Open Science Framework deposit cited above therefore timestamps
 208 the record but is retrospective, and we make no claim of prospective regis-
 209 tration on it. Two consequences follow. An external record does not exclude
 210 undisclosed post-hoc analytic choices; and the two analyses we ourselves iden-
 211 tify as post-hoc (the acceleration-factor extraction of Table 6 and the venue
 212 classification of Section 4.5) are labelled as such wherever they appear.

213 *Search strategy.* Seven bibliographic databases (PubMed/MEDLINE, IEEE
 214 Xplore, Scopus, Web of Science, SpringerLink, OpenAlex, and Semantic
 215 Scholar) were searched between 23 and 24 July 2026, with no publication-date
 216 limit applied; the 1995–2026 span quoted elsewhere in this article is the pub-
 217 lication range of the studies that were included, not a restriction imposed on
 218 the search. Queries were executed through each database’s API rather than
 219 its web interface, so the complete Boolean match set was retrieved for every
 220 term combination rather than a ranked sample, explaining why the initial
 221 identification count is large. The strategy combined three required concept
 222 groups: neuroimaging (Block B1), reconstruction and acceleration (Block B2),
 223 and safety, failure modes, and evaluation outcomes (Block B3), with MeSH
 224 headings and wildcard truncation adapted per database (Supplementary Ta-
 225 ble S5). Table 1 details the executed query structure and hit breakdown across
 226 databases.

227 The breadth of this search is by design. Evidence of reconstruction failure
 228 is rarely a paper’s stated subject: artifacts, hallucinations and metric–reader
 229 disagreements are most often reported in passing inside studies whose declared
 230 contribution is a new architecture, so a query tuned narrowly to safety vocabu-
 231 lary would miss much of this evidence. Clinical and engineering literatures also
 232 use distinct vocabularies (sequences and artifacts against inverse problems, un-
 233 rolled optimization and generative priors), and neither database family indexes
 234 the other completely. We therefore tuned for sensitivity rather than precision

Table 1: Query structure executed in each database and record hit breakdown.

Database	Executed Query Structure	Controlled Vocabulary	Fields Searched	Hits (N)
PubMed / MEDLINE	(B1 _{ft} [TIAB] OR B1 _{mesh}) AND (B2 _{ft} [TIAB] OR B2 _{mesh}) AND (B3 _{ft} [TIAB] OR B3 _{mesh}) NOT (X[TIAB])	Yes: 9 MeSH headings OR-ed into the three blocks	Title/Abstract + MeSH	24,693
SpringerLink	(B1) AND (B2) AND (B3) NOT (X); quoted phrases, no truncation operators	No	Full record	1,598
IEEE Xplore	(B1) AND (B2) AND (B3) NOT (X); wildcard truncation retained	No	Metadata	778
Scopus	TITLE-ABS-KEY(B1) AND TITLE-ABS-KEY(B2) AND TITLE-ABS-KEY(B3) AND NOT TITLE-ABS-KEY(X)	No	TITLE-ABS-KEY	127
Semantic Scholar	(B1) + (B2) + (B3), where denotes OR and + denotes AND	No	Title/Abstract	66
OpenAlex	title_and_-abstract.search:B1, title_and_-abstract.search:B2, title_and_abstract.search:B3	No	Title/Abstract	34*
Web of Science	TS=(B1) AND TS=(B2) AND TS=(B3) NOT TS=(X)	No	Topic (TS)	31
Total identified records				27,327
Concept blocks combine as (B1) AND (B2) AND (B3) NOT (X). B1 neuroimaging: "brain MRI", "cerebral MRI", neuroimag*, DWI, FLAIR, T1w, T2w, SWI; MeSH <i>Brain, Magnetic Resonance Imaging, Neuroimaging</i>. B2 AI and accelerated reconstruction: "MRI reconstruction", "accelerated MRI", "undersampled reconstruction", "k-space reconstruction", "unrolled network", GAN, "diffusion prior"; MeSH <i>Deep Learning, Machine Learning</i>. B3 safety, artifacts and evaluation: hallucinat*, "lesion suppression", reconstruction artifact*, PSNR, SSIM, "reader study"; MeSH <i>Artifacts, Diagnostic Errors, Reproducibility of Results</i>. X exclusion: "animal model", "case report", "in vitro". * denotes truncation. Complete term lists, MeSH headings and per-database syntax are given in Supplementary Table S5. Searches were executed between 2026-07-23 and 2026-07-24, with no publication-date limit. *OpenAlex returned 50 raw hits; 34 passed record-completeness filtering.				

and accepted the screening burden. The funnel from 27,327 records to 263 studies is the intended signature of that choice: the burden falls on initial screening, which reviewers can absorb, rather than on coverage, which no downstream step could recover.

239 *Eligibility and selection.* Records were managed in the review’s own Python
240 pipeline rather than a commercial screening platform; the scripts, the per-record
241 decision files and the per-database query logs are released, so each step below
242 is re-executable. Deduplication by exact identifier matching (PMID/DOI) and
243 string/TF-IDF similarity (Jaro-Winkler ≥ 0.97 , cosine ≥ 0.95) removed 403
244 duplicates (1.47%), leaving 26,924 unique records. That rate is low enough to
245 suggest a matching failure but is structural: PubMed’s MeSH arm returned
246 24,693 records and the six engineering arms 2,634 between them, and sets that
247 barely overlap leave little to deduplicate. This is not a claim of completeness,
248 and one failure is on record: peer review identified an arXiv preprint and its
249 journal version that both survived to the included set, carrying different DOIs,
250 no shared PMID, and titles differing by terminal punctuation, which was be-
251 low the thresholds above. It was removed at revision, taking the corpus from
252 264 reports to 263 studies (Section 4). Automated matching will not catch
253 a preprint–journal pair whose metadata diverges in this way, and the honest
254 inference is that a residual duplicate rate of the same order cannot be excluded.

255 Three reviewers (D.T.M., T.V.P., T.N.T.D.) screened records against the
256 pre-specified PICOS criteria (Table 2) at both stages, under a primary-screen-
257 plus- adjudication design: each record received one primary label of *include*,
258 *exclude*, *flagged* or *background only*, and every record labelled *flagged* went to
259 a second reviewer for adjudication. Labelling was deliberately conservative,
260 so any record with an unclear PICOS dimension, or whose text we could not
261 obtain, was flagged rather than excluded. Title and abstract screening advanced
262 1,149 records to full-text assessment, of which 109 were labelled *include* on the
263 abstract alone and 1,040 *flagged*; both labels carried the record forward, and
264 the distinction records only whether the abstract was sufficient to decide. A
265 further 25,573 were excluded and 202 set aside as background. Full texts were
266 pursued through institutional and title-level library subscriptions, open-access
267 repositories, preprint servers, PubMed Central and inter-library request until
268 no route remained: 1,068 of the 1,149 (93.0%) were obtained and assessed, and
269 the 81 that could not be obtained are broken down by recorded reason with the

270 selection results (Section 4).

271 The three reviewers divided the 1,068 obtained reports between them un-
272 der the same primary-screen-plus-adjudication design used at title and abstract:
273 each report received one primary full-text label, and a report was escalated to
274 a second reviewer wherever the primary label was *flagged*, wherever an inclu-
275 sion decision rested on a borderline PICOS dimension, or wherever the reviewer
276 requested a second reading. Escalated reports were resolved by consensus dis-
277 cussion between the two assessors, with the third reviewer consulted where the
278 two did not converge. The escalation volumes this produced are reported below,
279 and the per-report labels, escalation marks and reasons are released in the de-
280 cision ledger. We state the design in this form because it determines what can
281 be computed from the ledger: one primary label per report means no chance-
282 corrected inter-rater coefficient is recoverable from full-text screening, which is
283 why the separate criterion-reproducibility exercise below was run. In total, 263
284 primary studies met all inclusion criteria upon full-text verification (two ad-
285 ditional reports were excluded as duplicate publications). Full-text exclusions
286 ($n = 785$) were categorized by pre-specified PICOS criteria: wrong intervention
287 ($n = 326$), out of scope ($n = 256$), wrong outcome ($n = 98$), wrong population
288 ($n = 93$), wrong study design ($n = 4$), non-English ($n = 1$), and non-primary
289 articles ($n = 7$). A further 18 reports were retained as background material
290 rather than as included studies, so the 1,068 assessed reports resolve as 263
291 included, 785 excluded on PICOS grounds, 2 excluded as duplicate publications
292 and 18 retained as background. The complete decision ledger is released as
293 **fulltext_screening_decisions.csv**.

294 *Publication year.* Every analysis partitioning the corpus by time uses the
295 year of the version of record as indexed by the publisher (the issue year for
296 journal articles, the proceedings year for conference papers) rather than first
297 online posting or preprint deposit. The rule matters at era boundaries, and we
298 adopt the indexed year because it is what a reader can verify from the reference
299 list, Crossref and PubMed without online-first metadata, which is unevenly
300 available across the 87 venues represented. The same value is recorded in the

301 `year` field of every bibliography entry and in the `publication_year` column of
 302 `included_characteristics.csv`, so the era counts in Table 9 are recomputable
 303 from either. One consequence should not be read as an inconsistency: the
 304 searches ran on 23–24 July 2026, yet 41 of the 263 included studies (15.6%)
 305 carry a 2026 issue year, being articles published online ahead of print before the
 306 searches ran but assigned by the publisher to a 2026 issue.

307 *Screening agreement, and what it does and does not cover.* The primary
 308 screen described above records one label per record, so no chance-corrected co-
 309 efficient is recomputable from those decisions alone. We therefore ran a separate
 310 duplicate-screening exercise. A random sample of 214 records already assessed
 311 at title and abstract was re-screened independently by all three reviewers, blind
 312 both to the recorded decision and to one another. Reviewers judged the four
 313 PICOS dimensions from title, journal and abstract, opening the DOI link only
 314 where the abstract was insufficient, and returned one of three labels per record:
 315 include, exclude or unsure. The sample was drawn without stratification; the
 316 drawn records, the three label sets and the agreement computation are released,
 317 so the sample is auditable record by record. All 642 cells were completed and
 318 no record was dropped for missingness.

319 Across all 214 records Fleiss’ $\kappa = 0.877$ (95% bootstrap CI 0.823–0.925;
 320 10,000 resamples, seed 20260821), with observed agreement $\bar{P} = 0.933$ against
 321 $\bar{P}_e = 0.455$ expected by chance. The three reviewers agreed unanimously on
 322 194 of 214 records (90.7%), and pairwise Cohen’s κ ran from 0.845 to 0.931.
 323 One sensitivity analysis is worth reporting because it is the one a reader is most
 324 likely to run. Restricting to the 193 records on which no reviewer recorded
 325 *unsure* raises Fleiss’ κ to 0.943 (95% CI 0.900–0.979). That complete-case figure
 326 discards 21 records, and it discards precisely the records at least one reviewer
 327 could not decide, so it overstates agreement on the population the coefficient
 328 is meant to describe. We report 0.877 as the estimate and 0.943 only as the
 329 value obtained when the hard cases are removed. Two qualifications travel with
 330 that figure and should not be dropped. First, it measures agreement on the
 331 screening decision, not on the extraction labels from which every prevalence

count in this article is derived; those remain unaudited for reliability, and the only error rate we can quote for them is the measured 4/61 (6.6%) on the design field (Section 5.2). Second, the conservative labelling policy routed almost the whole full-text workload to adjudication in any case (1,040 of the 1,149 records advanced at title and abstract were flagged rather than included outright), so a high coefficient is evidence that the criteria were applied consistently, not that the borderline records were easy to classify.

What the primary decision records themselves support, and what we therefore also report, are the escalation volumes that policy produced. At full text, 84 reports were escalated to a second reviewer for adjudication and 29 of those were included; a further 77 reports carry a mandatory human-re-inspection mark in the released files, of which 40 were included after that review. Screening recall is bounded by the search strategy and by the criteria as applied rather than estimated against an independent reference set of known-eligible records, since no such set exists for this question; Section 5.2 treats both points as limitations. The duplicate-screened sample, both label sets and the agreement computation are released as `kappa_human_results.csv`, alongside the full decision ledger in `fulltext_screening_decisions.csv`.

Data extraction. Reviewers extracted study characteristics using the pre-specified form (Supplementary Table S6), capturing method family, reported failure modes, evaluation paradigm, field strength, pulse sequence, reference standard and reproducibility indicators, with a verbatim supporting quotation for every assigned value. Method, failure-mode and evaluation categories were assigned from each record’s title, abstract and controlled vocabulary against the vocabulary fixed in advance (Supplementary Table S5); that deliberately conservative scope is why every prevalence count here is a lower bound rather than an exhaustive tally. Field strength and pulse sequence, typically stated only in acquisition subsections, were additionally taken from full text where available. We did not contact investigators: a variable not stated in the source was recorded as *not reported* and never inferred, which is why the 54 studies without a stated field strength divide into 50 whose full text does not state it

Table 2: PICOS elements and their application in this review.

Element	Definition	Application in this Review
Population	Target imaging modalities, clinical domains, and scanner physics	Human brain MRI acquisitions (acute ischemic stroke, neurovascular lesions, intracranial haemorrhages, tumours, and structural T1w, T2w, FLAIR, DWI scans across 1.5T, 3T, and 7T field strengths).
Intervention	AI-based and accelerated reconstruction methods, and their safety/fidelity evaluation	Deep learning and accelerated reconstruction methods (unrolled physics-based networks, GANs, diffusion priors, self-supervised, denoising, super-resolution), together with methods for evaluating their fidelity, artifacts, robustness, or failure modes.
Comparison	Baseline metrics, observers, and reference standards	Conventional pixel-wise metrics (PSNR, SSIM, NRMSE), perceptual/learned metrics (LPIPS, FID), model observers, fully sampled reference images, or radiologist reader assessments.
Outcome	Reconstruction quality, fidelity, and failure characterization	Image-quality/fidelity measures, artifact and failure-mode characterization, metric-radiologist agreement, uncertainty estimates, robustness to distribution shift, and diagnostic/observer outcomes.
Study Design	Methodological criteria and empirical validation settings	Primary empirical studies, algorithmic benchmarking, counterfactual interventional stress-testing, and human or model observer evaluations.

and 4 assessed on abstract or partial text.

Quality appraisal, and the reader/algorithmic partition. Methodological quality was evaluated with three design-matched instruments: QUADAS-2 (Whiting et al., 2011) across its four domains for diagnostic accuracy and radiologist reader studies; a reporting-transparency check against STARD principles (Bossuyt et al., 2015), reported descriptively rather than scored, since STARD yields no risk-of-bias rating; and, for algorithmic benchmarking studies, a reproducibility and robustness appraisal covering code and data availability, use of a named public benchmark, whether the reference standard was a fully sampled acquisition, and whether robustness to acquisition or distribution shift was reported. ROBINS-I (Sterne et al., 2016) was not applied, since few included studies are the non-randomised interventional designs it was built for.

The partition between the two instruments is not the one the extraction labels originally implied, and the correction matters for every reader-share statis-

377 tic here. Extraction recorded 61 studies as observer or reader designs; three-
 378 reviewer appraisal of their full text found four containing no human reader at
 379 all, each using the language of observer evaluation for an automated surrogate.
 380 Those four were reclassified as algorithmic, so **the verified reader subset is**
 381 **57 and the algorithmic set is 206**, and $57 + 206 = 263$ closes the corpus
 382 exactly. All 57 were appraised by all three reviewers across two rounds, 39 in
 383 the first and 18 in the second, so every QUADAS-2 tally rests on 57 studies and
 384 $57 \times 7 = 399$ domain judgements. Every reader-derived count in this article
 385 uses 57, and the released per-study file encodes the same partition with the re-
 386 classification recorded per study. One further study (S001) carries an appraisal
 387 note that a human reader could not be *confirmed*, only abstract-level text being
 388 available; it is retained in the reader subset, since non-ascertainment is not a
 389 finding of absence, and it is flagged in the deposit.

390 *Appraisal agreement.* The three reviewers rated every applicable domain
 391 Low, Unclear or High independently, each judgement carrying a verbatim sup-
 392 porting quotation, before joint review to harmonize thresholds. Five first-round
 393 studies were later re-appraised on recovered full text (below), so their 35 judge-
 394 ments no longer rest on three independent ratings and are excluded from the
 395 coefficient. Over the remaining 52 studies and 364 judgements Fleiss' $\kappa = 0.788$
 396 (95% bootstrap CI 0.739–0.833; 10,000 resamples, seed 20260821). That figure
 397 averages two rounds that behaved differently, and the difference is more infor-
 398 mative than the average. The 34 first-round studies retaining three indepen-
 399 dent ratings were unanimous on all 238 of their judgements, so they contribute
 400 $\kappa = 1.00$ by construction and carry no information about latent reliability. The
 401 18 of the second round were unanimous on 60 of 126 (47.6%), and across those
 402 126 Fleiss' $\kappa = 0.390$ (95% CI 0.293–0.484), falling in the fair range on the
 403 Landis–Koch bands, with an interval reaching into moderate. The procedure
 404 was identical across rounds; the subsets were not, the second round compris-
 405 ing studies the first pass deferred as not straightforward. Two readings remain
 406 open: that those studies report their methods less completely and are genuinely
 407 harder to rate, and that the first round was less independent in practice than

408 in design ; these data do not separate them.

409 The disagreement has a structure worth recording rather than smoothing.
 410 Across the 126 second-round judgements one reviewer rated Low 33.3% of the
 411 time and another 61.9%, a difference of threshold rather than of reading, which
 412 is why κ sits at 0.390 while raw agreement is 63.5%. Six judgements split
 413 three ways with no majority; all six were recorded as *Unclear*, on the ground
 414 that a panel unable to converge is itself evidence the report did not support
 415 a confident judgement, which is what *Unclear* encodes in QUADAS-2. The
 416 rule was applied to those six and no other case, and the per-reviewer labels are
 417 released so a reader can apply a different rule and recompute. Section 5.2 states
 418 what this does and does not establish. Table 3 summarizes corpus-level quality
 419 indicators and the QUADAS-2 domain judgments.

Table 3: Corpus-level provenance and methodological quality indicators across the 263 included studies, with QUADAS-2 domain-wise risk of bias and applicability concerns for the full appraised diagnostic/reader subset ($n = 57$, rated across seven domains; independently by all three reviewers except for the five studies re-appraised on recovered full text). The 39 studies of the first appraisal round and the 18 of the second are pooled here; the remaining 206 studies were appraised on reproducibility and robustness instead ($57 + 206 = 263$). Six of the 399 domain judgements split three ways with no majority and are recorded as *Unclear* (Section 3).

Indicator / Domain	Value / Risk of Bias Breakdown
Publication period	1995–2026; 229/263 (87.1%) published in 2020 or later, and 174/263 (66.2%) in 2023 or later, indicating a predominantly recent evidence base.
Basis of assessment, current	Full text available for 258/263 (98.1%); partial or truncated full text for 1/263 (0.4%); title, abstract and MeSH only for 4/263 (1.5%).
Basis of assessment, at appraisal	The first appraisal round was made on the text obtainable at the time, which for 18/263 studies was abstract and MeSH only (6 of them within the QUADAS-2 subset). A later library sweep recovered full text for 14 of those 18 (including 5 of the 6), and all 14 have since been re-appraised on the recovered full text, so appraisal basis now stands at 257/263 full text, 2/263 partial text and 4/263 abstract and MeSH only (within the QUADAS-2 subset, 56/57 on full text and 1/57 [S001] on abstract-level text). The two rows now differ by a single study (S200), appraised on truncated text whose full version has since become available.

(Table 3 continued)

Leading venues	The eight most frequent of the 87 distinct venue titles represented in the corpus, accounting for 128/263 (48.7%) of included studies: <i>Magnetic Resonance in Medicine</i> (40), <i>IEEE Transactions on Medical Imaging</i> (17), <i>AJNR</i> (15), <i>Magnetic Resonance Imaging</i> (14), <i>JMRI</i> (12), <i>Scientific Reports</i> (12), <i>NeuroImage</i> (9), <i>Medical Image Analysis</i> (9).
Predominant design	Algorithmic / methodological reconstruction studies ($n = 206$), alongside a verified observer/reader subset ($n = 57$) (Radmanesh et al., 2022; Lang et al., 2024). Extraction recorded 61 reader designs; appraisal of their full text found 4 with no human reader, and those 4 were reclassified as algorithmic ($202 + 4 = 206$; $61 - 4 = 57$). Of the 57: 34 are appraised and verified by all three reviewers; 18 were appraised by all three in a second round and await final adjudication of 6 split domain judgements; and 5 (S004, S020, S027, S034, S115) were re-appraised by the reviewers on full text recovered after the first round, with domain judgements verified against the source text. Every reader-derived statistic in this article uses 57.
Reference standard	Expert reader consensus (70), an unspecified stated ground truth (62), a fully sampled acquisition (69), a conventional or standard-of-care acquisition (15); not stated in 47/263 (17.9%).
Reproducibility	Code availability stated in 97/263 (36.9%); a named public dataset or benchmark used in 109/263 (41.4%), most commonly fastMRI, HCP, ADNI and IXI. Availability is variable and remains an area for improvement.
Sampling and setting	Sampling frame not stated in 95/263 (36.1%); recruitment centre not stated in 174/263 (66.2%). Ethics approval or informed consent reported in 159/263 (60.5%).
QUADAS-2 Risk of Bias ($n = 57$ Reader / Diagnostic Accuracy Studies)	
Patient Selection	Low: 15/57 (26.3%); Unclear: 19/57 (33.3%); High: 23/57 (40.4%). The weakest domain: sampling is predominantly purposive or pathology-enriched, and most of the appraised studies state no enrolment rule.
Index Test (AI Reconstruction)	Low: 26/57 (45.6%); Unclear: 9/57 (15.8%); High: 22/57 (38.6%). Low risk falls below half once the second round is included. High ratings arise where readers scored accelerated reconstructions with the reference visible, or where test parameters were set using reference-derived labels.
Reference Standard	Low: 26/57 (45.6%); Unclear: 15/57 (26.3%); High: 16/57 (28.1%). High ratings arise where no artifact-free reference exists, or where lesion truth is defined by the comparator images themselves (incorporation bias). Reading the recovered full text moved four of the re-appraised studies from Low to High here, each because the comparator images, or software applied to them, supplied the truth against which they were scored.

(Table 3 continued)

Flow and Timing Low: 45/57 (78.9%); Unclear: 7/57 (12.3%); High: 5/57 (8.8%). The strongest domain, since retrospective undersampling of the same raw k -space gives a zero interval between index test and reference.

QUADAS-2 Applicability Concerns ($n = 57$)

Patient Selection	Low: 36/57 (63.2%); Unclear: 4/57 (7.0%); High: 17/57 (29.8%), chiefly healthy-volunteer, lesion-free, or non-clinical cohorts.
Index Test	Low: 30/57 (52.6%); Unclear: 4/57 (7.0%); High: 23/57 (40.4%), where the evaluated method is image-domain post-processing, cross-field synthesis, harmonization, or a quality index rather than reconstruction from under-sampled k -space.
Reference Standard	Low: 39/57 (68.4%); Unclear: 3/57 (5.3%); High: 15/57 (26.3%).

Across all 228 risk-of-bias judgements: 112 (49.1%) Low, 50 (21.9%) Unclear, 66 (28.9%) High. Across all 171 applicability judgements: 105 (61.4%) Low, 11 (6.4%) Unclear, 55 (32.2%) High. Over the 364 judgements that rest on three independent ratings Fleiss' $\kappa = 0.788$ (95% CI 0.739–0.833), but agreement is not uniform across the two appraisal rounds: the 238 retained first-round judgements were unanimous, the 126 of the second round reach $\kappa = 0.390$ (95% CI 0.293–0.484); the 35 judgements of the 5 re-appraised studies are excluded, having been conducted subsequently on recovered full text. Section 3 gives the decomposition and Section 5.2 what it bounds. The high proportion of Unclear ratings reflects unreported blinding and sampling procedures, compounded by the one first-round study for which only abstract-level text is available. Six second-round judgements split three ways with no majority and are recorded as Unclear under the rule stated in Section 3. Per-study ratings, each with the verbatim supporting quotation, are released as `included_characteristics_supplement.csv`.

420 *Derived venue classification.* The analysis in Section 4.5 groups studies by
421 publication community. That grouping is post-hoc, was not part of the fixed
422 protocol, and is stated in full in Supplementary Table S13 so it can be re-
423 produced or contested. Each study's journal title, and nothing else about the
424 study, is matched case-insensitively against three ordered keyword lists (engi-
425 neering and methods, then general neuroscience and neuroimaging, then clinical
426 imaging), with anything unmatched left *unclassified* rather than forced into a
427 group. The ordering is load-bearing, since titles such as *Medical Image Anal-*
428 *ysis* would otherwise match two lists, and it is applied identically to all 263
429 studies. The rule as executed and the resulting per-study assignment are re-
430 leased as `5_Code/venue_classification.py` and `venue_community_classi-`
431 `fication.csv`.

432 *Synthesis.* Due to heterogeneity in acquisition protocols, reconstruction ar-
433 chitectures, datasets, and outcome measures, evidence was synthesised narra-

tively in accordance with SWiM guidelines (Campbell et al., 2020). Studies were structured around the primary research questions (RQ1–RQ3) and grouped by reconstruction method family, field strength, pulse sequence, and failure mode. Quantitative values reflect named primary studies rather than pooled meta-analytic estimates; no effect measure is pooled, and no meta-analytic model is fitted. Percentages are reported to one decimal place and rounded half-up throughout, so 9/80 prints as 11.3% and not 11.2%. Sensitivity analysis was performed by recalculating headline prevalence counts on the 258 studies verified against complete full text (Section 5.2).

4. Results

All counts and prevalence statements in this section describe the 1,068 reports assessed in full and the 263 included studies; because 81 candidate reports remain unretrievable, they are lower bounds on the literature rather than estimates of it.

4.1. Selection and corpus

From 27,327 records identified across seven databases, multi-tier deduplication removed 403 duplicates, leaving 26,924 unique records screened at title and abstract. Of these, 1,149 proceeded to full-text eligibility, and full texts were obtained for 1,068 of them (93.0%; Section 3). Each of the 81 reports still out of reach was checked against its publisher platform and assigned a recorded reason: 53 sit behind title-level subscriptions our institution does not hold, across nineteen publishers; 13 are Springer conference chapters sold only per chapter; 4 are ISMRM proceedings abstracts issued without a PDF; 4 resolve to a whole proceedings volume or to front matter rather than a single report; and 7 carry no DOI and no locatable source. Supplementary Table S14 gives the per-publisher breakdown. Access difficulty is itself informative, though it bounds the missing evidence rather than characterising it. Of the 341 assessed reports obtainable only through title-level subscriptions and equivalent barriers,

294 (86.2%) proved ineligible on the same PICOS criteria, 2 were retained as background and 45 were eligible ($294 + 2 + 45 = 341$; the 45 resolve to 44 studies, one being a preprint adjudicated a duplicate). Raw non-retrieval counts therefore overstate the missing evidence by roughly sevenfold. But those 45 were concentrated in subscription clinical journals, and Section 4.5 shows they differ systematically from the open literature in exactly the variable this review measures, so 86.2% is an upper bound on what non-retrieval costs us, not a demonstration that it costs us nothing. Of the 1,068 assessed reports, 787 were excluded (785 on PICOS grounds, 2 as duplicate reports of included studies) and 18 recorded as background, leaving **263 included primary studies** spanning 1995–2026 (Fig. 1). Each included study was verified against the fullest text obtainable for it: complete full text for 258, partial text for 1, and title, abstract and controlled vocabulary for 4 (Section 3). Their aggregate characteristics are given in Table 4, and their thematic distribution across the research questions is summarized in Supplementary Table S7; all 263 are listed, with full references, in Supplementary Tables S1–S3.

Corpus growth is recent: 34 studies appeared in 1995–2019, 55 in 2020–2022 and 174 in 2023–2026.

4.2. RQ1: the metric–reader comparison is almost never made, and where readers appear they agree only partially

Across the 84 studies reporting objective fidelity metrics and the 57 verified reader studies, the comparisons that actually exist show agreement between fidelity metrics and radiologist assessment that is partial and condition-dependent (Table 5). The size of that evidence base should be stated in the text and not left to the table caption: only two of the four rows in Table 5 were collected on reconstructions of undersampled k -space. Tang et al. (2025) collected radiologist rankings on synthetically corrupted image pairs, and Sommer et al. (2020) on clinical scans corrupted by real patient motion rather than by undersampling. The table that carries RQ1 therefore rests on two on-target rows, which is the honest size of the direct evidence for the metric–reader comparison

Table 4: Characteristics of the 263 included studies.

Demographic / Method Dimension	Category Breakdown	(n)	Percentage / Key Methodological Notes
Publication Period	1995–2019	$n = 34$	12.9% (Classical CS/PI & early CNNs)
	2020–2022	$n = 55$	20.9% (Unrolled physics & GAN priors)
	2023–2026	$n = 174$	66.2% (Diffusion priors & 3D/4D foundation models)
Reconstruction Paradigm (multi-label; 339 labels recorded)	Compressed Sensing / Parallel Imaging	$n = 93$	Iterative sparsity & coil sensitivity encoding
	Generative (GAN/Diffusion/VAE)	$n = 80$	Adversarial perception & posterior sampling
	Motion Correction	$n = 57$	Intra-scan motion artifact suppression
	Physics-based Unrolled	$n = 37$	Explicit k -space data consistency layers
	Denoising	$n = 30$	Image-domain noise suppression
	Super-resolution	$n = 25$	Resolution enhancement post-acquisition
MRI Pulse Sequences (multi-label; 490 labels)	Self-supervised / Unsupervised	$n = 17$	Training without fully sampled reference scans
	T1-weighted (T1w)	$n = 163$	High-resolution anatomical cortical profiling
	T2-weighted (T2w)	$n = 122$	Ventricular & CSF pathology assessment
	FLAIR	$n = 85$	Hyperintense peri-ventricular lesion detection
	Diffusion-Weighted (DWI)	$n = 72$	Acute ischemic lesion (DWI core) evaluation
	SWI / T_2^* -weighted	$n = 48$	Punctate cerebral microbleed hypointensity voids
Magnetic Field Strength (single-label; predominant field where several are used)	3.0 Tesla (3T)	$n = 153$	58.2% (Primary clinical scanner benchmark)
	7.0 Tesla (7T)	$n = 35$	13.3% (Ultra-high-field research acquisitions)
	1.5 Tesla (1.5T)	$n = 21$	8.0% (Routine clinical throughput imaging)
	Not stated	$n = 50$	19.0% (Full text obtained; no field strength stated)
	Not ascertainable	$n = 4$	1.5% (Only abstract or partial text available)
Evaluation Paradigm (multi-label; 177 labels recorded across 144 studies)	Pixel-wise (PSNR/SSIM/NRMSE)	$n = 84$	47.5% of recorded (legacy fidelity reporting)
	Observer / Reader Studies	$n = 57$	32.2% of recorded (task-based reader assessment; verified subset)
	Uncertainty Estimation	$n = 18$	10.2% of recorded (posterior variance maps)
	Learned / Perceptual	$n = 18$	10.2% of recorded (rank-trained metrics)
Co-reporting. Of the 144 studies with a recorded evaluation paradigm, 28 report more than one. Critically, only 18 studies in the entire corpus (6.8% of 263; 12.5% of the 144 recorded) report <i>both</i> a pixel-wise fidelity metric <i>and</i> a reader assessment on the same data, which is the minimum precondition for computing metric-versus-reader agreement on which safety claims implicitly depend. See Section 4.			

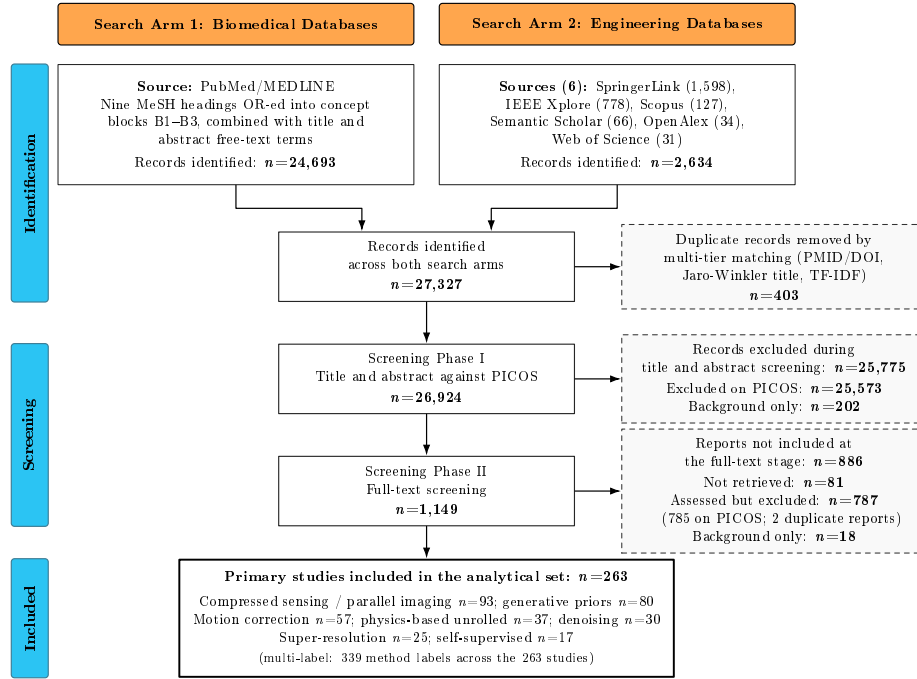


Figure 1: PRISMA 2020 flow diagram for study selection across the two search arms. Multi-tier deduplication reduced 27,327 identified records to 26,924 unique records; title and abstract screening advanced 1,149 candidates. Full texts were obtained for 1,068 of them (93.0%); 81 could not be retrieved. Following PRISMA 2020, exclusion and background retention are recorded as distinct dispositions: of the 1,068 reports assessed in full, 787 were excluded (785 on explicit PICOS grounds and 2 as duplicate reports of included studies) and a further 18 were retained as background material rather than excluded, leaving 263 primary studies in the analytical set ($263 + 787 + 18 = 1,068$).

492 in this setting. In the fastMRI 2020 challenge’s reader phase, six radiologists
493 ranked submissions for quality of pathology depiction; their concordance, by
494 Kendall’s coefficient, was 0.457 in the $4\times$ track and 0.386 in the $8\times$ track, rising
495 to 0.781 only in the transfer track, and the organisers report that concordance
496 improved as SSIM scores diverged, which is to say the metric separated methods
497 reliably only where their outputs already differed grossly (Muckley et al., 2021).
498 When radiologist preference is instead used as the training signal, purpose-built
499 quality networks predict expert rankings at $75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$,
500 against radiologists’ own interobserver agreement of $61.5\% \pm 5.5\%$; the same
501 study finds rank-based reading far more reliable than Likert scoring (70.4% ver-
502 sus 25% agreement between radiologists) (Tang et al., 2025). Reader tolerance

of acceleration is real but bounded: in a retrospective evaluation spanning accel-
erations up to $100\times$, 94% of volumes (65 of 69) were rated sufficient for potential
screening use at up to $14\times$ (interreader intraclass correlation 0.875), with the
threshold for full diagnostic quality lower still (Radmanesh et al., 2022). The
one direct clinical observation of the failure this review is about comes from an
emergency-department and inpatient comparison, and it is a single case. Across
66 paired examinations in which a two-minute ultrafast protocol (T1w, T2/T2*,
FLAIR and DWI, machine-learning-assisted reconstruction) was read against a
ten-minute reference protocol, two blinded neuroradiologists agreed on the main
clinical diagnosis in 65 of 66 cases (98.5%), rated overall diagnostic quality no
different between protocols ($P > .05$), preferred the reference protocol for image
noise and geometric distortion ($P < .05$) and preferred the accelerated protocol
for reduced motion artifact ($P < .01$) (Lang et al., 2024). The single discrepant
case is the one that matters here: a punctate focus of restricted diffusion, read
as possible acute-to-subacute infarct on the reference DWI, was less conspicu-
ous on the accelerated DWI, which the authors attribute to a combination of
image-quality difference and section positioning, the lesion being smaller than
the section thickness (4 mm reference against 5 mm accelerated) and so subject
to partial-volume contamination. They draw the inference themselves, that the
protocol’s sensitivity to small findings is an open question. One case in 66 is
not a prevalence estimate and we do not offer it as one. It is a worked instance
of the pattern this review predicts: an aggregate quality comparison that finds
no difference, and inside it a focal finding that a reader saw and the aggregate
did not. Where readers assess artifacts directly, blinded reading detects what
reference-based metrics cannot express: a two-neuroradiologist blinded study of
28 clinical scans with real patient motion found significant artifact-score reduc-
tions after network correction ($p < .03$), a setting in which the motion-corrupted
reference invalidates PSNR and SSIM by construction (Sommer et al., 2020).
Motion also interacts with the sampling scheme itself, appearing as ghosting
under Cartesian trajectories and as streaking or blurring under radial ones,
so the visual form of the same physical corruption differs by protocol (Schau-

man et al., 2026). The reference pipeline is not neutral either: conventional SWI homodyne phase filtering has been shown to create punctate artifactual microhaemorrhage-like foci, so processing can manufacture findings as well as remove them (Li et al., 2015).

No included study reports a rank correlation between a pixel-wise fidelity metric and radiologist diagnostic assessment for deep learning-based brain reconstruction, and none evaluated a model observer. The second statement needs a boundary that the first does not. Our search vocabulary (Table 1, block B3) contains reader-study and metric terms but no model-observer vocabulary, so a study that used a channelised Hotelling or non-prewhitening observer and described it only in those terms could have been missed at retrieval. What we can say is that none of the 263 studies retrieved, screened and extracted here reports one, and that the term does not appear in any extracted evaluation label; whether the true field-wide count is zero or merely very small, it is not a paradigm this corpus can be said to use. That absence is itself the RQ1 finding: only 18 of 263 studies (6.8%, a screening-level lower bound; Section 3) record a fidelity metric and a reader assessment on the same data, so the field’s central proxy relationship, that high fidelity scores imply diagnostic adequacy, is largely unmeasured rather than merely disputed.

Table 6 reports how evaluation practice varies with the acceleration factor actually stated in each study. It is built only from what the corpus records: the acceleration factor was extracted post hoc from the full texts of all 263 included studies with a verbatim supporting quotation for every value, and 63 studies (24.0%) state one. Studies are grouped by the highest factor at which they evaluate their method. The table deliberately carries no fidelity threshold, reader-score difference or metric–reader rank statistic per acceleration band, because no included study reports one: that absence is the finding of Section 4, not an omission from the table.

Table 5: What the reader studies in this corpus actually measured about metric–reader agreement (RQ1). Every value is quoted from the named study, with the reported coefficient and its definition stated. No included study reports a metric–reader rank correlation for deep learning-based brain reconstruction; the rows below are the closest measurements that exist, and the second column states for each row what the readers were actually shown, because two of the four rows were not collected on reconstructions of undersampled k -space.

Study	Readers and design	Comparison performed	Reported quantity (as defined in the study)
fastMRI 2020 challenge (Muckley et al., 2021)	6 radiologists ranking submissions for quality of pathology depiction	Radiologist concordance across accelerations; SSIM-based ranking vs reader ranking	Kendall's $W = 0.457$ ($4\times$), 0.386 ($8\times$), 0.781 (transfer track); concordance improved as SSIM scores diverged
Tang et al. (2025)	7 radiologists, rank-based reading; 2 also Likert scoring. Rankings were collected on synthetically corrupted image pairs, not on reconstructions of undersampled k-space	Quality networks trained on radiologist rankings vs MSE/SSIM as optimisation targets	Ranking accuracy $75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$ with reference image; radiologist interobserver agreement $61.5\% \pm 5.5\%$; ranking vs Likert agreement 70.4% vs 25%
Radmanesh et al. (2022)	Retrospective; neuroradiologist scoring of diagnostic quality at accelerations up to $100\times$	Reader-assigned acceleration thresholds for diagnostic and screening quality	$65/69$ volumes (94%) sufficient for potential screening at $\leq 14\times$; interreader ICC 0.875 ; mean absolute interreader difference in threshold acceleration factor $0.7\times$ (level 1) and $5.4\times$ (level 2)
Sommer et al. (2020)	2 neuroradiologists, blinded, 28 clinical cases with real motion	Artifact severity (5-point scale) before vs after network correction	Significant reduction in mean artifact scores ($p < .03$); reference-based metrics inapplicable, as the reference itself is motion-corrupted

Table 6: Evaluation practice by the highest acceleration factor (R) each study reports, among the 63 of 263 included studies that state one. Bands are half-open, so a study reporting exactly $R = 4$ falls in the second band and one reporting exactly $R = 8$ in the third. Counts are studies; percentages are within band. *Reader*, any observer or reader assessment recorded; *M+R*, a pixel-wise fidelity metric and a reader assessment recorded on the same data; *Halluc.*, hallucination or fidelity instability among the recorded failure modes; *Artifacts*, aliasing, ghosting or Gibbs artifacts among them. Acceleration factors were extracted from full text after the registered extraction round, each with a verbatim supporting quotation and excluding matches inside reference lists and citation contexts; the 407 retained quotations come from the 63 studies that state a factor, a mean of 6.5 quotations per such study, and are released as `acceleration_factors_evidence.csv`. The extraction is screening-level, so counts are lower bounds. No study at any acceleration factor reports a rank correlation between a fidelity metric and reader judgement.

Acc. (R)	Studies	Reader	M+R	Halluc.	Artifacts	Sequences most often evaluated
$2 \leq R < 4$	32	7 (21.9%)	1 (3.1%)	7 (21.9%)	25 (78.1%)	T1w 15, T2w 12, FLAIR 11
$4 \leq R < 8$	17	5 (29.4%)	1 (5.9%)	2 (11.8%)	12 (70.6%)	T1w 9, T2w 9, FLAIR 7
$R \geq 8$	14	5 (35.7%)	4 (28.6%)	2 (14.3%)	8 (57.1%)	T1w 9, T2w 8, FLAIR 8
All	63	17 (27.0%)	6 (9.5%)	11 (17.5%)	45 (71.4%)	T1w 33, T2w 29, FLAIR 26

The R -reporting subset is close to the whole corpus on reader assessment (27.0% against 21.7% for all 263), so it is not a systematically better-evaluated stratum. Reader assessment rises with acceleration (21.9% $\rightarrow$ 29.4% $\rightarrow$ 35.7%) and paired metric-reader reporting concentrates in the most accelerated band (4/14 against 1/32 in the $2 \leq R < 4$ band), but the counts are small and most studies in every band (including 9 of the 14 with $R \geq 8$) report no reader assessment at all. Artifact description is the most frequently recorded failure mode at every acceleration level, whereas hallucination or fidelity instability is recorded in 11 of 63.

4.3. RQ2: four mechanisms explain the divergence

Four mechanisms account for why the metrics miss what the readers see. Table 7 states each one formally alongside the clinical failure it permits, and Supplementary Table S8 gives the fuller derivations and evidence sources.

Spatial averaging. Conventional L_2 metrics average residual error uniformly over all $H \times W \times D$ voxels, so the total decomposes over lesion and background regions:

$$\begin{aligned} \text{MSE}(\hat{X}, X) &= \frac{V_{\text{lesion}}}{V_{\text{total}}} \text{MSE}_{\text{lesion}} \\ &\quad + \left(1 - \frac{V_{\text{lesion}}}{V_{\text{total}}}\right) \text{MSE}_{\text{bg}}. \end{aligned} \quad (1)$$

With lesion volume fraction $f = V_{\text{lesion}}/V_{\text{total}}$ and local error ratio $r = \text{MSE}_{\text{lesion}}/\text{MSE}_{\text{bg}}$, the global PSNR change caused by a localized edit follows

Table 7: The four mechanisms by which a whole-volume fidelity score fails to register a focal diagnostic error (RQ2). The third column names a failure the mechanism permits, not one this review has observed at a stated frequency; the fourth states why the score does not move. Numerical entries are computed from Equation (2) under the stated error model, not measured in any primary study.

Mechanism	Formal statement	Clinical failure it permits	Why the score does not move
(a) Spatial error averaging	L_2 error is averaged uniformly over all $H \times W \times D$ voxels, so a lesion contributes with weight $f = V_{\text{lesion}}/V_{\text{total}}$	Erasure of an acute lacunar infarct on DWI or FLAIR, or of a punctate microhaemorrhage on SWI or T_2^*	By (2), erasing a 100 mm^3 infarct at $r = 100$ moves global PSNR by $0.027\text{--}0.036 \text{ dB}$; a 5 mm^3 microhaemorrhage by $1.3\text{--}1.8 \times 10^{-3} \text{ dB}$
(b) Reference dependence	Every reference-based score presupposes a registered, artifact-free reference image	A valid motion-corrected reconstruction is penalised against a corrupted reference, and genuine correction is scored as error	The comparison is no longer between reconstruction and truth; PSNR and SSIM become uninterpretable rather than merely noisy
(c) Loss symmetry	$(\hat{X}_i - X_i)^2 = (X_i - \hat{X}_i)^2$: the objective is indifferent to the sign of the error	A suppressed hyperintense lesion is scored identically to added background noise of the same magnitude	Diagnosis is asymmetric where the metric is not: a missed finding and a spurious brightening carry the same penalty
(d) Realism–reward trade-off	Adversarial and perceptual objectives reward outputs typical of the training distribution, $\log D_\phi(\hat{X})$, not outputs faithful to the measurement	Synthesis of plausible vessels, sulci or texture that was never acquired	Perceptual scores can rise as measurement fidelity falls, so the objective pays for the hallucination it should penalise

571 directly:

$$|\Delta\text{PSNR}| = 10 \log_{10} (1 + f(r - 1)) \quad (\text{dB}). \quad (2)$$

572 The consequence is quantitative rather than rhetorical, and it is worth anchoring
573 on a lesion a neuroradiologist would expect to resolve rather than on the smallest
574 arithmetically convenient one. An acute lacunar infarct of 100 mm^3 is a sphere
575 5.8 mm across: on a clinical DWI acquired at $4\text{--}5 \text{ mm}$ section thickness (Lang
576 et al., 2024) that is several voxels wide in-plane and spans one to two sections, so
577 it is resolved, if only just, in the through-plane direction. It occupies a volume
578 fraction $f = 6.3\text{--}8.3 \times 10^{-5}$ of a $1.2\text{--}1.6 \text{ L}$ brain. Even at a hundredfold local
579 error ratio ($r = 100$), Equation (2) gives $|\Delta\text{PSNR}| = 0.027\text{--}0.036 \text{ dB}$. Erasing

580 a resolvable acute infarct outright therefore moves global PSNR by about three
 581 hundredths of a decibel, which is the same order as the third decimal place of a
 582 score conventionally reported to two.

583 The limiting case is worth stating precisely, because it is easy to state loosely.
 584 A punctate cerebral microhaemorrhage of 5 mm^3 is 2.1 mm across. It is *not*
 585 resolved on clinical DWI, whose voxels are roughly $12\text{--}20 \text{ mm}^3$, so no recon-
 586 struction can be said to erase it there; it is not resolved in the first place. It
 587 is several voxels across on the high-resolution SWI or T_2^* -weighted acquisition
 588 on which microhaemorrhages are actually read, and on that sequence its era-
 589 sure costs $f = 3.1\text{--}4.2 \times 10^{-6}$ and $|\Delta\text{PSNR}| = 1.3\text{--}1.8 \times 10^{-3} \text{ dB}$, below the
 590 precision at which the score is reported at all. Scaling in the other direction, a
 591 500 mm^3 cortical infarct costs $\approx 0.15 \text{ dB}$ ($0.13\text{--}0.18 \text{ dB}$ across the same brain-
 592 volume range). Fig. 2 plots the identity across this range, with the sub-voxel
 593 region marked. The argument does not need the extreme case: it holds by one
 594 to two orders of magnitude at a lesion size that is unambiguously resolved and
 595 unambiguously matters. These are computed bounds from the stated f and r ,
 596 not measurements from any primary study, and the error model behind them, a
 597 uniform background error and a free local error ratio, is an assumption we state
 598 rather than a property of any reconstruction.

599 *Loss-function symmetry.* MSE and SSIM are indifferent to the sign of the
 600 error, since $(\hat{X}_i - X_i)^2 = (X_i - \hat{X}_i)^2$, whereas diagnosis is not: suppressing a
 601 true hyperintense lesion means the finding is missed, while adding background
 602 noise costs nothing clinically. An evaluation respecting that asymmetry must
 603 penalise the two directions differently, for example

$$\begin{aligned} \mathcal{L}_{\text{asym}}(\hat{X}, X) = & \alpha \sum_{i \in \Omega_{\text{lesion}}} \max(0, X_i - \hat{X}_i)^2 \\ & + \beta \sum_{i \in \Omega_{\text{bg}}} (\hat{X}_i - X_i)^2, \quad (\alpha \gg \beta). \end{aligned} \quad (3)$$

604 Several included studies already build objectives of this shape: lesion-mask-
 605 weighted losses motivated explicitly by the dilution of lesion gradients among
 606 background voxels (Lee et al., 2026a), overlap-weighted reconstruction losses

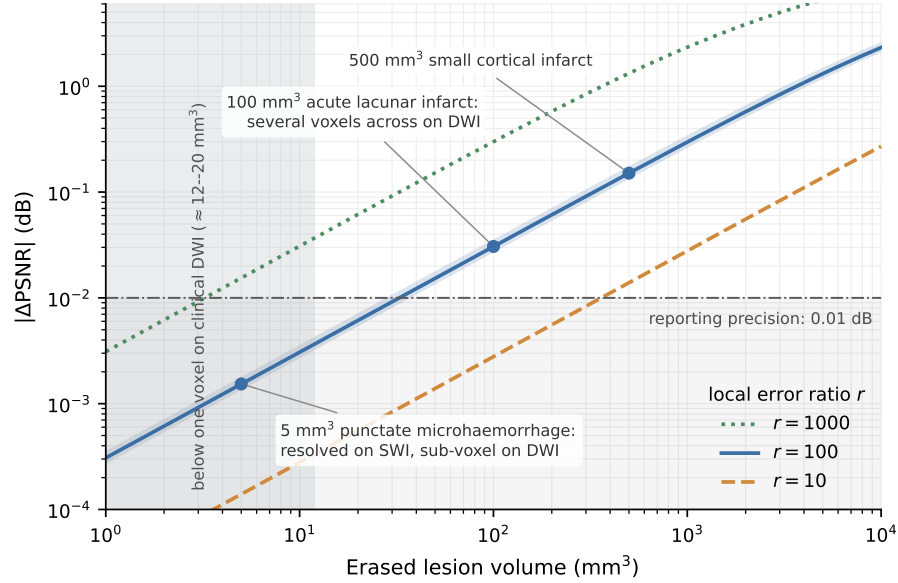


Figure 2: The PSNR cost of erasing a focal lesion, from Equation (2). Curves give $|\Delta\text{PSNR}|$ against erased lesion volume for three local error ratios r at a 1.4 L brain; the shaded band around $r = 100$ spans the 1.2–1.6 L range. Markers show the three clinical anchors used in the text. The horizontal line at 0.01 dB is the two-decimal precision at which PSNR is conventionally reported, and the shaded left-hand region is the volume below a single voxel on a clinical DWI acquisition, where erasure is not defined because the finding is not resolved. Every value is computed from the identity under the stated error model; nothing in this figure is measured.

607 that up-weight structure against background (Jayakumar et al., 2023), and
 608 gradient-based structure-preservation terms (Zhang et al., 2026). Their exis-
 609 tence is evidence that the symmetry problem is recognised in practice, though
 610 none of these studies pairs the correction with a reader assessment.

611 *Realism–reward trade-off.* Adversarial and perceptual objectives reward out-
 612 puts that look like the training distribution, which is not the same as outputs
 613 faithful to the measurement, so a generative reconstruction can gain plausible
 614 texture that was never acquired (Zhou et al., 2024; Bhadra et al., 2021).

615 *Reference dependence.* All four metrics presuppose a registered, artifact-free
 616 reference. Intra-scan motion invalidates that assumption, and 69 of 263 studies
 617 rest their reference standard solely on a fully sampled acquisition, which is valid
 618 only where the subject did not move.

619 *4.4. RQ3: what current evaluation paradigms can and cannot establish*

620 Read against those mechanisms, the corpus supports a paradigm-by-
 621 paradigm account of what each evaluation approach can demonstrate about
 622 prior-induced diagnostic error. Table 8 states the account compactly; the
 623 argument behind it is given here, because two of its six rows are analytical
 624 claims we derive rather than findings we extracted, and an analysis presented
 625 only in a table has not been argued.

626 *What the four empirical paradigms establish.* Pixel-wise fidelity metrics es-
 627 tablish aggregate agreement with a reference and nothing about focal change:
 628 by (2) the penalty for erasing a resolvable acute infarct is of the order of the
 629 reporting precision, so a method can pass any threshold anyone would set while
 630 failing on the finding that matters. Perceptual and rank-trained metrics track
 631 reader preference better than MSE or SSIM as an optimisation target, but they
 632 inherit the variability of the readers they were trained on and, in the one study
 633 of this kind in the corpus, were calibrated on synthetically corrupted images
 634 rather than on reconstructions of undersampled k -space (Tang et al., 2025), so
 635 their transfer to the reconstruction setting is assumed rather than shown. Un-
 636 certainty estimates flag where a model is unsure, which is not the same question:
 637 a prior confident enough to remove a lesion is confident in the region where it
 638 removed it, so posterior variance can be low exactly where the diagnostic error
 639 is. Task-based observer assessment does settle the question directly, and it is
 640 the only paradigm here that does, which is why its scarcity in this corpus (57 of
 641 263 studies, and 18 paired with a metric) is the review’s central finding rather
 642 than an incidental one.

643 *Two limits that follow from the physics, not from the corpus.* The first
 644 concerns the check that reconstruction papers most often use to argue that an
 645 output is faithful to the data: the measurement-space residual $\|\mathbf{A}\hat{X} - Y\|_2 / \|Y\|_2$
 646 against a tolerance. This check cannot detect prior-induced edits at all, and the
 647 reason is structural rather than statistical. Write the reconstruction as the sum
 648 of its measured and unmeasured components, $\hat{X} = \mathbf{A}^+ \mathbf{A} \hat{X} + (\mathbf{I} - \mathbf{A}^+ \mathbf{A}) \hat{X}$. The
 649 second term lies in the null space $\mathcal{N}(\mathbf{A})$, and $\mathbf{A}$ annihilates it by definition, so

any amount of structure written there, an erased lesion or an invented vessel
 alike, leaves the residual exactly unchanged. Undersampling is what creates that
 null space; the more aggressive the acceleration, the larger the subspace in which
 a learned prior can write freely without moving the number the field uses to
 certify data consistency. A residual check is therefore a two-sided measurement-
 fidelity band and not a safety test, and its apparent reassurance grows least
 trustworthy precisely where acceleration is highest.

The second limit is the constructive corollary. If prior-supplied content lives
 in $\mathcal{N}(\mathbf{A})$, then isolating $(\mathbf{I} - \mathbf{A}^+\mathbf{A})\hat{X}$ and inspecting it is the natural place to
 look for erasure and hallucination, since that is where both must reside (Bhadra
 et al., 2021; Gottschling et al., 2025). This is the one paradigm in Table 8 that
 is not represented in the corpus at all, and its practical obstacle is the same one
 that limits third-party auditing generally: computing the projection requires the
 forward operator, hence the raw k -space and the coil sensitivities, which com-
 mercial inline pipelines do not export (Section 5). Causal and counterfactual
 auditing, which asks the most direct question of all, whether a specific finding
 survives a controlled change of acquisition, appears in a small part of the cor-
 pus and almost never for physics-constrained k -space reconstruction (Pawlowski
 et al., 2020; Güngör et al., 2023; Peng et al., 2026).

The answer to RQ3 is therefore not that current evaluation is weak but
 that it is mismatched: each paradigm in use answers a question adjacent to the
 safety question, and no paradigm now in use establishes reconstruction safety
 on its own. The two that could, task-based observer assessment and null-space
 inspection, are respectively rare and absent, and the requirements in Section 4.7
 follow from that pairing.

4.5. Structure of the evaluation gap

The counts above establish that safety-relevant evaluation is scarce. Analysis
 of the same per-study records establishes where the scarcity comes from, and
 indicates a division of labour rather than uniform neglect.

Table 8: What each evaluation paradigm can and cannot establish about prior-induced diagnostic error (RQ3). Rows 1–4 summarize evaluation practice and performance as reported in the included studies; quoted values are representative results from the cited primary studies, not pooled estimates. Rows 5–6 state analytical limits derived by the authors from the acquisition physics (our own analysis; Section 4) and are labelled as such rather than attributed to any primary study.

Evaluation Paradigm	Core Mechanism	What It Can and Cannot Establish	Key References
1. Pixel-wise fidelity metrics	L_2 / SSIM whole-image voxel averaging	Aggregate agreement with a reference only. Cannot resolve focal change: by (2), erasing a resolvable 100 mm^3 lacunar infarct at $r = 100$ costs 0.03 dB. Requires a registered reference	Recorded in 84 of the 144 evaluation-labelled studies
2. Perceptual & learned metrics	Networks trained on radiologist rankings	Tracks reader preference better than MSE/SSIM as an optimisation target ($75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$ against readers' own $61.5\% \pm 5.5\%$). Inherits training-reader variability, and was calibrated on synthetically corrupted data rather than clinically undersampled acquisitions, so transfer is assumed (Section 4.4)	Tang et al. (2025) ($n = 18$ in corpus)
3. Uncertainty estimation	Posterior sampling / Bayesian variance $p(X Y)$	Flags out-of-distribution inputs and regions of low confidence. Does not attribute missing structure to prior vs sampling operator; high confidence in hallucinated regions is possible	Khawaled and Freiman (2024) ($n = 18$ in corpus)
4. Task-based observer assessment	Human reader performance on a diagnostic task; model observers are its automated form, unrepresented here (Barrett et al., 1993; Barrett and Myers, 2004; International Commission on Radiation Units and Measurements, 1996)	Establishes diagnostic sufficiency directly; the gold standard in assessment theory. Costly and reader-variable ($61.5\% \pm 5.5\%$, falling to 25% under absolute Likert scoring), and rarely paired with a metric: only 18/263 report both	Radmanesh et al. (2022), Lang et al. (2024) ($n = 57$ in corpus)
5. Measurement-space residual check	$\ \mathbf{A}\tilde{X} - Y\ _2 / \ Y\ _2$ against tolerance	Cannot detect prior-induced edits: the residual is invariant to anything in $\mathcal{N}(\mathbf{A})$. A two-sided measurement-fidelity band, not a safety test	Common diagnostic check in the reviewed studies
6. Null-space / hallucination mapping	Structure written into the null-space component $(\mathbf{I} - \mathbf{A}^+ \mathbf{A})\hat{X}$	Isolates prior-supplied content, where erasure and hallucination must reside. Requires $\mathbf{A}$, hence raw k -space and coil sensitivities	Bhadra et al. (2021), Gottschling et al. (2025); formalization for reconstruction auditing is author-derived

679 *The benchmarks cannot show the failure the field is worried about.* One hun-
 680 dred and nine of the 263 studies (41.4%) name a public dataset, and across
 681 the whole corpus those references resolve to eleven datasets: fastMRI, the Hu-
 682 man Connectome Project, ADNI, IXI, UK Biobank, BraTS, OASIS, Calgary–
 683 Campinas, M4Raw, the developing Human Connectome Project and TCGA.
 684 Dataset labels are multi-valued (136 cohort assignments across the 109 studies),
 685 and of the 109, the cohorts named are healthy volunteers or population sam-
 686 ples (55, 50.5%), chronic neurodegenerative cohorts (31, 28.4%), brain tumour
 687 cohorts (16, 14.7%), or raw- k -space benchmarks not curated for pathology (34,
 688 31.2%). These four shares are multi-label and sum to 124.8%, a study naming
 689 two cohorts being counted in both; every other percentage here is single-label.
 690 *Within this corpus, no study benchmarks against a dataset curated for acute is-*
 691 *chemic stroke or intracranial haemorrhage.* The sharper version of this finding
 692 sits in the annotation layer the field has already built. fastMRI+ supplies 7,570
 693 expert bounding-box annotations across thirty brain pathology categories (Zhao
 694 et al., 2022), and not one of the thirty is a haemorrhage category of any kind,
 695 microhaemorrhage included. The apparent counterexample should be stated
 696 plainly rather than left for a reader to find, because it is the category closest
 697 to this article’s own worked example: fastMRI+ *does* carry an image-level *La-*
 698 *cunar Infarct* label (113 annotations across 32 subjects), alongside study-level
 699 *Small Vessel Chronic White Matter Ischemic Change* and a *Global Ischemia*
 700 label recorded for a single subject. It does not rescue the benchmark, and the
 701 reasons are the ones that matter for reconstruction safety. No fastMRI+ cat-
 702 egory distinguishes acute from chronic infarction, so a lacunar label carries no
 703 acuity and cannot identify the acute presentation whose erasure is at issue. The
 704 limit is in the source images as much as in the label set: the annotated se-
 705 ries are axial T2-FLAIR, T1w or contrast-enhanced T1w only, so neither DWI,
 706 on which infarct acuity is established, nor SWI or T_2^* -weighted imaging, on
 707 which microhaemorrhage is read, was annotated at all. Acute vascular pathol-
 708 ogy is therefore absent even where annotation infrastructure exists, and it is
 709 absent for a reason that adding labels alone would not repair. Supplementary

710 Table S10 sets out the full derivation, including the sub-selection of 1,001 of
711 the 5,847 fastMRI brain studies, the category listing, and the single-annotator
712 limitation that travels with these annotations. Stroke-lesion datasets do exist,
713 ISLES (Hernandez Petzsche et al., 2022) and ATLAS (Liew et al., 2022) among
714 them, but they are segmentation benchmarks distributed as processed images
715 without raw k -space, so they cannot serve reconstruction evaluation as released.
716 This matters mechanically: lesion erasure cannot be measured on cohorts that
717 contain no acute lesions, so a model can lead every public leaderboard in the
718 field while remaining untested against the failure mode that would matter most
719 in an emergency department. Where a pathology-labelled dataset is used at all
720 it is BraTS, whose tumours are large, high-contrast mass lesions, close to the
721 easiest possible case for lesion preservation and the least informative about the
722 lacunar and punctate findings at issue here. The benchmark infrastructure and
723 the metric therefore fail in the same direction.

724 *Rigour is divided between two literatures.* Of the 263 studies, 84 (31.9%)
725 release code but report no reader assessment, 44 (16.7%) report a reader assess-
726 ment but release no code, 122 (46.4%) do neither, and only 13 (4.9%) do both.
727 The division tracks publication venue. Applying the post-hoc classification rule
728 stated in Section 3 to all 263 venues, clinical imaging journals ($n = 131$) carry
729 reader assessment at nearly three times the rate of engineering and methods
730 venues ($n = 78$), which in turn release code at nearly twice the clinical rate;
731 general neuroscience and neuroimaging venues ($n = 37$) sit between. Supple-
732 mentary Table S12 gives the full cross-tabulation. The rule leaves 17 of the 263
733 journal titles unmatched, and those are reported as a fourth *unclassified* group
734 rather than forced into one of the three; the four groups between them account
735 for all 57 reader studies and all 97 code releases. The tendency for the two
736 practices to fall in different studies rather than the same ones gives a Fisher
737 exact odds ratio of 0.43 ($p = 0.013$). The estimate is insensitive to the one
738 labelling choice that could move it: taking the reader subset as the 61 studies
739 originally labelled at extraction rather than the 57 verified at appraisal gives an
740 odds ratio of 0.43 and $p = 0.010$. We nonetheless report the association descrip-

tively rather than as a hypothesis test, because the venue grouping it rests on is a post-hoc keyword rule that was not part of the fixed protocol, and a p -value computed on a partition chosen after seeing the data does not carry its nominal meaning. The direction and magnitude are what we rely on. The magnitude is the part that does not depend on the choice: computational verifiability and diagnostic verifiability are largely held by different communities, only one study in twenty holds both, and a safety argument requires both.

Awareness of a failure mode does not produce the evaluation that would detect it. Generative studies record hallucination or fidelity instability in 31/80 (38.8%) of cases, the highest share among the four largest method families, against 15.1% for compressed-sensing and parallel-imaging work and 12.3% for motion correction. Yet at 9/80 (11.3%) they are also among the two least reader-evaluated of those families, while the small self-supervised group records the corpus’s highest hallucination share (8/17, 47.1%) and no reader assessment at all (0/17); per-family shares are in Table 9 and Fig. 3. The sub-fields that most clearly recognise that their models can synthesise structure are the least likely to run the test capable of detecting it.

Evaluation is not broadening as the field grows. Reader-assessment share fell from 32.4% (1995–2019) to 18.2% (2020–2022), recovering only partially to 20.7% (2023–2026) while the corpus grew fivefold (Table 9 and Fig. 4). What recovery there is comes entirely from the subscription-walled clinical literature: among 2023–2026 studies obtainable only through title-level library subscriptions, reader assessment runs at 52.0% (13 of 25), against 15.4% (23 of 149) in the openly accessible remainder of that era. Reader evidence, where it exists, sits disproportionately behind paywalls, and a review that cannot retrieve those texts will understate it; Section 5.2 takes the direction of that bias seriously rather than dismissing it. Among the 144 studies recording any evaluation paradigm, four in five record exactly one, and the share recording two or more fell across the same periods (Supplementary Table S12). Of the 84 studies recording a pixel-wise metric, 60 (71.4%) record no reader, uncertainty or perceptual measure beside it.

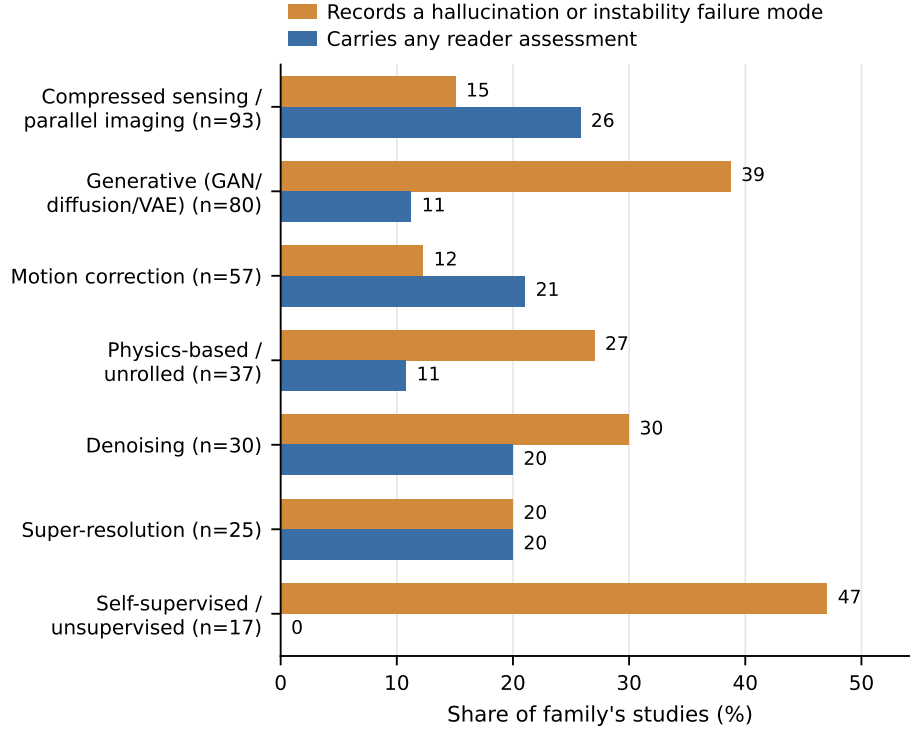


Figure 3: Reader assessment against hallucination awareness, by method family (recorded extraction labels; lower bounds). Bars are the share of each family’s studies that record a hallucination or instability failure mode (orange) and the share that carry any reader assessment (blue), using the verified reader subset ($n = 57$). Bar labels are rounded to the nearest integer; the one-decimal values quoted in the text are the same quantities (for example 15 against 15.1%, 39 against 38.8%). Generated from the released per-study file (`included_characteristics.csv`) by `5_Code/make_figures.py`; every value is recomputable from the deposit.

772 *Readers are not displaced away from the sequences where the stakes are*
773 *highest.* One expectation is not borne out. The comparison must be made
774 on disjoint strata to mean anything, because pulse sequence is a multi-label
775 field and the naive contrast double-counts the 107 studies that acquire both
776 an acute-pathology and a structural sequence (Supplementary Table S11 gives
777 the correction). That the two disjoint strata below sum to $23 + 84 = 107$, the
778 same number, is a third coincidence of unrelated tallies rather than a tran-
779 scription error, verified against the released per-study file as the two flagged
780 in Section `efsec:requirements` are. Restricted to studies acquiring one group

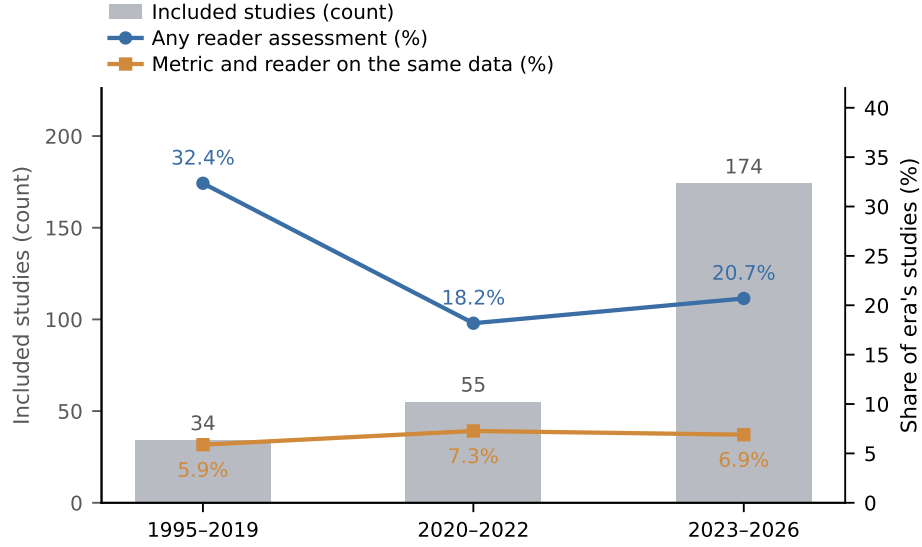


Figure 4: Evaluation practice across publication eras. Grey bars give the number of included studies per era (34, 55, 174); lines give the share with any reader assessment and the share co-reporting a fidelity metric and a reader assessment on the same data, using the verified reader subset ($n = 57$). What recovery there is in 2023–2026 comes from subscription-walled clinical journals (52.0% reader assessment in that stratum against 15.4% in the open remainder; Section 5.2). Generated from the released per-study file by `5_Code/make_figures.py`; counts are lower bounds from recorded extraction labels.

781 and not the other, reader assessment runs at 5/23 (21.7%) for DWI or FLAIR
782 without T1w or T2w against 10/84 (11.9%) for T1w or T2w without DWI or
783 FLAIR, representing a difference in the direction one would hope for, resting
784 on 23 studies in the smaller stratum and not approaching significance (Fisher
785 exact odds ratio 2.06, $p = 0.31$). We report it as descriptive and draw only
786 the negative conclusion it supports: there is no evidence that reader assessment
787 is *displaced away* from the sequences on which acute pathology is read. The
788 deficit is one of scarcity and of distribution across research communities, not of
789 misdirection within the clinical literature.

790 4.6. Methodological quality and risk of bias

791 Across all 57 QUADAS-2 appraised diagnostic and reader studies, method-
792 ological quality was highest for Flow and Timing (45/57 low risk of bias, 78.9%),
793 reflecting the standard retrospective undersampling paradigm where reconstruc-

Table 9: Evaluation practice across publication eras and method families (recorded labels; lower bounds). Reader-assessment share fell as the corpus grew and recovers only partially in the latest era, and that recovery is carried by subscription-walled clinical journals. Among the four largest method families the generative group, the one most associated with hallucination, is one of the two least reader-evaluated and is indistinguishable from physics-based unrolled work; the small self-supervised family ($n = 17$) records no reader assessment at all. Reader shares use the verified reader subset ($n = 57$; Section 3), so counts sum to 57 across the era rows and are multi-label across the family rows.

Era	n	Reader	Uncert.	Co-report
1995–2019	34	32.4% (11/34)	5.9%	5.9%
2020–2022	55	18.2% (10/55)	5.5%	7.3%
2023–2026	174	20.7% (36/174)	7.5%	6.9%
Method family (multi-label)				
Family	n	Reader share		
Compressed sensing / parallel imaging	93	25.8% (24/93)		
Motion correction	57	21.1% (12/57)		
Super-resolution	25	20.0% (5/25)		
Denoising	30	20.0% (6/30)		
Generative (GAN/diffusion/VAE)	80	11.3% (9/80)		
Physics-based / unrolled	37	10.8% (4/37)		
Self-supervised / unsupervised	17	0.0% (0/17)		

tion and reference share the same acquisition timepoint without temporal degradation. Reference Standard and Index Test both sit at 26/57 low risk (45.6%), each now below half, with high ratings arising where radiologists scored accelerated images with reference ground truth visible or where lesion truth was defined by the comparator images themselves (incorporation bias). Patient Selection remains the weakest domain (15/57 low risk, 26.3%; 19/57 unclear, 33.3%; 23/57 high risk, 40.4%), as cohorts were predominantly retrospective, single-centre, and pathology-enriched without explicit prospective enrolment rules. Applicability concerns follow a similar distribution, with low concerns in Patient Selection for 36/57 (63.2%), Index Test for 30/57 (52.6%), and Reference Standard for 39/57 (68.4%).

The two rounds do not place the weaknesses in quite the same order, and the pooled shares above differ from the first round’s in every one of the seven domains. Low-risk shares fell for Index Test (53.8% to 45.6%), applicability of the Index Test (59.0% to 52.6%), Flow and Timing (82.1% to 78.9%), Reference Standard (46.2% to 45.6%) and applicability of Patient Selection (64.1% to 63.2%); they rose for Patient Selection risk of bias (23.1% to 26.3%) and applica-

bility of the Reference Standard (59.0% to 68.4%). The two largest movements concern the index test and the applicability of the reference standard, and the first of these runs against the appraised studies. The second-round studies are therefore not a favourable addition to the appraisal, which is why the pooled figures are the ones reported here. The first-round shares quoted here already incorporate the five studies re-appraised on recovered full text.

Corpus-wide reporting transparency evaluated against STARD principles revealed substantial omissions across the broader 263 studies: 109/263 (41.4%) provided no identifiable reference standard (47 stating none at all, Table 3, and a further 62 naming a ground truth they left unspecified), scanner field strength was unreported in 54/263 (20.5%), and 166/263 (63.1%) did not make code openly available. Overall, 220/263 studies (83.7%) omitted at least one key methodological indicator (field strength, reference standard, or open code). Two directions must be kept apart here, and conflating them has caused confusion elsewhere in this literature. With respect to the *reconstruction methods*, this reporting and selection pattern is anti-conservative: retrospective single-centre evaluation on curated data flatters reported performance relative to what prospective clinical deployment would show. With respect to *this review’s own conclusion*, the same pattern is conservative: if published performance is flattered and failure is under-reported, then the scarcity of safety-relevant evaluation we document is an understatement, and the true gap is at least as large as the one measured. The safety argument turns on the second direction and is not weakened by the first. Detailed domain-level judgments and supporting quotations are available in the open project repository.

4.7. Requirements for safety-oriented evaluation

Read together, the synthesis implies five requirements that any evaluation intended to establish reconstruction safety must meet, each stated with the corpus evidence that motivates it and none met by more than a small minority of included studies. **R1, focal task-based measurement:** every global fidelity score must be accompanied by a region-of-interest or task-based observer

841 measure, because spatial averaging is the first mechanism by which lesion-scale
 842 error disappears (60/84, or 71.4%, of pixel-metric studies record nothing beside
 843 the metric; this equals the artifact share of the *R*-reporting subset in Table 6
 844 (45/63) to the decimal, which is a coincidence of two unrelated tallies rather than
 845 a transcription error, verified against the released per-study file). **R2, paired**
 846 **metric–reader reporting**: metric validity claims require a reader comparison
 847 on the same data (6.8% of studies provide one). **R3, reference-independent**
 848 **validity**: evaluation must remain interpretable when no registered artifact-free
 849 reference exists (26.2% rest solely on a fully sampled acquisition). **R4, robust-**
 850 **ness under acquisition shift as a primary outcome**, reported as a result
 851 rather than a caveat; robustness or distribution shift is recorded as a failure-
 852 mode concern in 76/263 studies (28.9%), almost always in the discussion rather
 853 than as a pre-specified endpoint. **R5, hallucination-specific detection**, op-
 854 erating on the image side, since measurement-space residual checks are provably
 855 blind to null-space edits. To these the analysis in Section 4.5 adds a precondition:
 856 benchmarks containing the pathology whose erasure is at issue, without
 857 which R1, R2 and R5 cannot be exercised at all.

858 5. Discussion

859 Three conclusions follow. First, the comparison on which the field’s central
 860 proxy rests is almost never made: no included study reports a rank correlation
 861 between a pixel-wise fidelity metric and radiologist diagnostic assessment,
 862 and only 18 of 263 record the two on the same data at all. Where readers
 863 do appear, what is measured is mostly agreement between readers, and it is
 864 weak and condition-dependent. The finding therefore rests more on the absence
 865 of measurement than on the measurements that exist. Second, the reason is
 866 structural, through the four mechanisms of Section 4: background voxels average
 867 focal error away, the loss functions are sign-symmetric where diagnosis is
 868 not, adversarial terms pay for invented texture, and every one of these metrics
 869 assumes a registered reference that motion destroys. The mechanisms are math-

870 ematical properties of the metrics themselves; how often each matters clinically
871 is the part the thin reader evidence leaves less certain. Third, no evaluation
872 paradigm now in use settles the safety question by itself, a conclusion resting
873 on the paradigm-by-paradigm synthesis of Section 4.

874 The analysis of where the gap comes from, which was not part of the fixed
875 protocol and is reported as exploratory throughout, changes what would follow
876 from these findings if it holds. The shortfall does not look like a uniform over-
877 sight; it co-occurs with two structural features. Evaluation practice is divided
878 between a community that releases code and a community that runs readers,
879 with only 4.9% of studies doing both (an association we interpret descriptively
880 in Section 4.5, its venue partition being post-hoc), and the benchmark datasets
881 named in this corpus contain no acute pathology, so the failure mode of great-
882 est clinical consequence is not merely under-measured but unmeasurable on the
883 datasets in common use. If that association is causal, which this design cannot
884 establish, recommendations addressed to individual authors will not close the
885 gap; what would be required is benchmark construction and reporting conven-
886 tions that force the two halves of the evidence together.

887 This matters because accelerated reconstruction has left the research setting.
888 Deep-learning acceleration, denoising and image enhancement ship inside com-
889 mercial MRI products, several of them cleared as software as a medical device:
890 one such tool, an FDA-cleared vendor-agnostic DICOM-domain enhancement
891 product, is evaluated in this corpus by a prospective multicentre multireader
892 trial (Bash et al., 2021), which is also a reminder that a cleared product can
893 be validated on one sequence family and deployed across others. Published
894 guidance in both jurisdictions increasingly asks manufacturers to demonstrate
895 performance for the device’s stated intended use and under conditions repre-
896 sentative of deployment rather than on curated development data: the FDA,
897 Health Canada and MHRA guiding principles on good machine learning practice
898 ask that datasets be representative of the intended patient population and that
899 testing be independent of training (U.S. Food and Drug Administration et al.,
900 2021), the EU AI Act imposes accuracy and robustness obligations on high-risk

901 AI systems (European Parliament and Council of the European Union, 2024),
 902 and the Medical Device Regulation requires clinical evidence appropriate to the
 903 device’s intended purpose (European Parliament and Council of the European
 904 Union, 2017). Our reading of that guidance is that aggregate pixel-wise fidelity
 905 cannot support a claim about small-lesion detection, because (2) shows the ag-
 906 gregate does not move when the small lesion is removed. That reading is ours:
 907 no regulator has reviewed or endorsed it, and none of the three instruments
 908 names an image-quality metric.

909 A practical obstacle compounds it, and it is the reason the requirements in
 910 Section 4.7 are addressed to infrastructure rather than to authors. Commer-
 911 cial inline pipelines export DICOM images after undisclosed normalisation, coil
 912 combination and vendor post-processing, and the raw complex k -space typi-
 913 cally never leaves the scanner. Every method in Table 8 that could detect a
 914 prior-induced edit needs something the exported DICOM does not carry: the
 915 null-space projection needs the forward operator and the coil sensitivities, a
 916 counterfactual audit needs the ability to re-run the acquisition under a con-
 917 trolled change, and even a measurement-space residual needs Y . Independent
 918 third-party auditing of a deployed reconstruction is therefore not merely difficult
 919 but structurally impossible without an open raw-data interface, and no amount
 920 of reporting discipline by authors of research papers can supply one.

921 Three things follow that are worth stating as claims rather than as recom-
 922 mendations.

923 *The cheapest useful change costs no new data.* A reconstruction paper that
 924 claims its method preserves small lesions can state, from (2) and its own brain
 925 volume and assumed local error ratio, the change in global PSNR that erasing
 926 the smallest lesion it claims to preserve would produce. For a lacunar infarct
 927 that is about 0.03 dB, which is smaller than the difference the paper is likely to
 928 be reporting between its method and its baseline. This is arithmetic, not a new
 929 experiment, and it converts an implicit claim into an explicit one at the point
 930 where the claim is made. Nothing in this review suggests such a line would be
 931 difficult; the finding is that nobody currently reports it.

932 *The two-community split is self-reinforcing, which is why exhortation has*
 933 *not worked.* Each community’s evaluation is close to the best it can do with
 934 the data it can obtain. Engineering venues evaluate on the datasets that carry
 935 raw k -space, and those contain healthy volunteers, chronic cohorts and tumours
 936 rather than acute pathology, so a reader study run on them would not test lesion
 937 preservation even if it were run. Clinical venues have the pathology, the readers
 938 and the acute presentations, and no access to raw k -space or to the vendor’s
 939 reconstruction, so a reproducibility claim is not available to them. The 4.9% of
 940 studies doing both are not more diligent than the rest; they are the studies that
 941 happened to sit where both kinds of access coincide. An instruction to individual
 942 authors to run more reader studies asks most of them to do something the data
 943 available to them makes uninformative.

944 *The missing object is a dataset, and its requirements are now specifiable.*
 945 The public resources discussed in this review supply each necessary compo-
 946 nent separately, and the distinction between what the corpus uses and what
 947 merely exists is part of the point: raw multi-coil k -space is available in three
 948 of the eleven datasets the included studies actually name (fastMRI, Calgary–
 949 Campinas, M4Raw); acute cerebrovascular pathology with expert segmentation
 950 exists in ISLES and ATLAS, which no included study uses because they are dis-
 951 tributed as processed images; and reader-level pathology annotation on brain
 952 MRI exists in fastMRI+, which no included study uses either. None combines
 953 them, and the combination is what every requirement in Section 4.7 presup-
 954 poses: raw k -space so the forward operator is known, acute lesions confirmed
 955 against a standard independent of the images being evaluated, lesion-level an-
 956 notation so a focal figure of merit can be computed, and a licence permitting
 957 redistribution so the benchmark can be used by people who did not collect it.
 958 That object does not exist, and until it does the safety question is not merely
 959 unanswered but unaskable on public data.

960 5.1. *Priorities for future research*

961 Three of the five requirements in Section 4.7 translate directly into research
962 priorities, and we do not restate them here: paired metric–reader reporting
963 (R2), robustness under acquisition shift reported as a primary endpoint rather
964 than a caveat (R4), and benchmarks carrying the pathology whose erasure is at
965 issue, which is the precondition on which R1, R2 and R5 all depend. Calibrated
966 voxel-level uncertainty that tracks local reconstruction error (Khawaled and
967 Freiman, 2024) belongs with R4 and is the form in which R4 is most likely to
968 be actionable for generative and self-supervised architectures.

969 A fourth priority does not follow from anything this review extracted, and we
970 separate it for that reason. Many current pipelines operate slice by slice and in-
971 troduce inter-slice discontinuities on isotropic 3D acquisitions, so reconstruction
972 and auditing frameworks should model 3D anatomical coherence directly (Peng
973 et al., 2026) while meeting scanner-side latency constraints. Slice-wise versus
974 volumetric processing was not an extraction field in this review. We state the
975 priority on mechanistic grounds and on the cited work, not on a corpus count.

976 5.2. *Threats to validity*

977 The strongest conclusions rest on a small subset of reader-based studies, and
978 we state that prominently rather than in passing: 57 of 263 studies carry any
979 reader assessment and 18 pair it with a metric. Four threat classes bound what
980 follows from that base.

981 *How to read the limitations below.* They are not all bounded in the same
982 way, and treating them as one undifferentiated list of caveats would misrepresent
983 what this review knows about itself. Three kinds appear. Some are *bounded*
984 *by released evidence*: the non-retrieval analysis, the paywall stratification and
985 the appraisal-agreement decomposition are each quantified from files a reader
986 can recompute. Some are *bounded by argument*: the direction of publication
987 bias and of the reporting asymmetry is reasoned rather than measured, and we
988 say which way each points. One is simply *unbounded*, and it is the sharpest:
989 extraction has no reliability estimate.

990 Three properties of the extraction layer follow from that and should be
 991 weighed together. First, its depth is deliberately shallow. Method, failure-
 992 mode and evaluation categories were assigned from title, abstract and controlled
 993 vocabulary, so the prevalence counts are floors on what studies *report at that*
 994 *level* rather than measurements of what they did. Second, the shallowness is
 995 visible in the counts themselves: only 84 of 263 studies are recorded as reporting
 996 a pixel-wise fidelity metric, and 39 of the 57 verified reader studies are recorded
 997 as reporting none, which for this literature is implausible as a description of
 998 practice and entirely plausible as a description of abstracts. A further 119 of
 999 263 studies (45.2%) carry no evaluation-paradigm label at all, which is why
 1000 we report both the /263 and the /144 denominators wherever the distinction
 1001 matters. Third, the headline 18/263 mixes two depths: the reader label was
 1002 verified against full text and corrected at appraisal, while the metric label was
 1003 not. That asymmetry biases the intersection downward by an unknown factor,
 1004 and it is the single respect in which our headline figure is least defensible as
 1005 a point estimate, though the direction is conservative, since correcting it could
 1006 only widen the measured gap.

1007 What we offer in place of a reliability coefficient is auditability, and the
 1008 substitution is deliberate rather than a shortfall we failed to notice. Every
 1009 extracted value is released with the verbatim quotation that supports it, so a
 1010 reader can check whether an assignment is *correct*, which is a stronger and more
 1011 useful guarantee than a coefficient showing that two people were *consistent*. It
 1012 is not a substitute for reliability in the statistical sense, and we do not claim
 1013 it is: consistency and correctness bound different failure modes, and only the
 1014 first is estimable from our records. A duplicate-extraction exercise on a random
 1015 subsample, reported with a κ , is the obvious next step and we name it as an
 1016 outstanding task rather than a completed one.

1017 *Construct validity.* We searched with broad, high-sensitivity terms for
 1018 architectures, artifacts and evaluation paradigms, and the corpus spans the
 1019 reconstruction-safety construct, from pixel-wise metrics through perceptual
 1020 losses to reader studies. What the corpus cannot guarantee is that a metric

1021 means the same thing wherever it appears: studies differ in what they treat as
1022 the quantity of interest, so a rank correlation with reader scores in one paper
1023 is not straightforwardly the same measurement as in another, and grouping
1024 such values under one heading is an interpretation we make, not a property of
1025 the data. The venue classification underlying Section 4.5 is likewise a post-hoc
1026 keyword assignment, released with the data rather than extracted from the
1027 studies.

1028 *Internal validity.* A protocol fixed before screening began, a documented
1029 search across seven databases, and independent screening and appraisal by three
1030 reviewers provide robust protection, and every extracted value is released with
1031 the verbatim snippet supporting it. Five specific threats bound these findings.
1032 First, 81 of the 1,149 candidate reports remain without obtainable text through
1033 structural licensing barriers. Screening the nearest obtainable stratum (the 341
1034 reports recovered only through title-level subscriptions) found 86.2% ineligible
1035 on the same criteria, so at that rate the 81 would contribute on the order of 11
1036 studies rather than 81. That bounds the missing evidence without establishing
1037 that the bias is nil. Non-retrieval is demonstrably *not* random with respect to
1038 the variable this review measures: within the same stratum, 2023–2026 stud-
1039 ies carry reader assessment at 52.0% against 15.4% in the openly accessible
1040 literature of that era. Both findings point the same way. Reader evidence is
1041 over-represented behind paywalls, so a review that cannot retrieve those texts
1042 understates how much reader assessment the field performs, and our headline
1043 scarcity figures are conservative bounds rather than point estimates. Second,
1044 four included studies were assessed on abstract and controlled vocabulary alone
1045 and one on partial text; a sensitivity analysis recomputing all headline metrics
1046 on the 258 studies verified against complete full text leaves every finding direc-
1047 tionally stable (e.g., metric–reader co-reporting remains 7.0% vs 6.8%). Third,
1048 every QUADAS-2 tally in this article rests on the whole verified reader subset
1049 ($n = 57, 399$ domain judgements), so none is computed on a partial base. The
1050 second round moved all seven domain distributions, and the movements that
1051 most concern the index test run against the appraised studies: low risk of bias

1052 for the Index Test falls from 53.8% to 45.6%, and high applicability concern
 1053 for the Index Test rises from 33.3% to 40.4% (Section 4). Four studies remain
 1054 abstract-only and are appraised on that basis. Fourth, reliability is measured
 1055 for screening and for appraisal but not for extraction, and extraction is the
 1056 sharpest methodological limitation of the review. The screening and appraisal
 1057 coefficients, and the decomposition that must be read with the appraisal figure,
 1058 are given in Section 3 and not restated here; the consequence for this section
 1059 is that 126 of the 399 QUADAS-2 judgements were reached at a level of agree-
 1060 ment a single rater could not be relied on to reproduce, so the domain shares
 1061 in Table 3 carry that uncertainty. It does not propagate to the prevalence find-
 1062 ings, which rest on extraction labels rather than QUADAS-2 ratings, but it
 1063 bounds how firmly the risk-of-bias picture can be stated. Neither screening nor
 1064 appraisal agreement transfers to extraction. The duplicate-screening sample is
 1065 also smaller than the 300–400 records a referee might reasonably ask for, and
 1066 the interval reflects that. Fifth, four studies recorded at extraction as observer
 1067 or reader designs were found at appraisal to contain no human reader and were
 1068 reclassified as algorithmic, moving the reader subset from 61 to 57 and the al-
 1069 gorithmic set from 202 to 206. The effect on the headline numbers is small and
 1070 in the direction of a larger gap: metric–reader co-reporting moves from 7.6%
 1071 to 6.8%, the 2023–2026 reader share from 23.0% to 20.7%, the code-and-reader
 1072 intersection from 5.3% to 4.9%, and the Fisher exact p from 0.010 to 0.013 at
 1073 an odds ratio that moves only from 0.43 to 0.43. Those four corrections are
 1074 also the source of the 4/61 (6.6%) error rate quoted above, and the design field
 1075 is the only one that received three-reviewer full-text verification: a comparable
 1076 rate elsewhere would not overturn any finding here, since the headline shares are
 1077 separated from their comparators by more than that margin, but it is a source
 1078 of variability measured in one place and assumed in the rest.

1079 *External validity.* Among the 209 studies stating a field strength the corpus
 1080 covers 1.5, 3 and 7 Tesla (21, 153 and 35 respectively), together with several
 1081 vendors’ coil configurations and the major pulse sequences; the remaining 54
 1082 (20.5%) state none, and nothing about field-strength coverage should be read

1083 into them. Tasks, protocols, architectures, datasets and metrics otherwise vary
 1084 substantially, which is why we report the direction and consistency of find-
 1085 ings rather than their magnitudes. Many algorithmic studies evaluate on retro-
 1086 spectively undersampled single-centre data, which flatters performance relative
 1087 to prospective multi-centre use; where the public data were themselves pre-
 1088 processed before undersampling, that inflation is larger again (Shimron et al.,
 1089 2022). Prospective multicentre reader evidence is thin, and the examples this re-
 1090 view leans on each carry a boundary that must travel with them: the emergency-
 1091 department comparison is single-centre and its one discrepant finding is a single
 1092 case (Lang et al., 2024); the widest-ranging acceleration reader study is retro-
 1093 spective and single-centre (Radmanesh et al., 2022); and the largest prospective
 1094 multicentre multireader trial evaluated 3D T1-weighted acquisitions only, with
 1095 an image-domain DICOM enhancement tool rather than reconstruction from un-
 1096 dersampled k -space, reported improved rather than degraded lesion conspicuity,
 1097 and has several authors affiliated with the vendor of the evaluated tool (Bash
 1098 et al., 2021); we cite it for what it does establish, that such tools are cleared
 1099 and deployed. Finally, the benchmark concentration of Section 4.5 (eleven pub-
 1100 lic datasets named across the corpus, not one curated for acute ischaemic stroke
 1101 or intracranial haemorrhage) caps how far any leaderboard result can be carried
 1102 into emergency imaging.

1103 *Conclusion validity.* Because pixel-wise metrics agree only loosely with ra-
 1104 diologist judgement (RQ1), a study resting on PSNR and SSIM alone supports
 1105 a safety claim less well than one with readers or uncertainty-aware observers,
 1106 and we weight the evidence accordingly. Publication bias favouring successful
 1107 reconstructions is a material limitation and works in the same direction, mak-
 1108 ing the reported prevalence of failure evidence an understatement: a literature
 1109 with little incentive to report failure is exactly the corpus a review about failure
 1110 must draw on, and the absence of a common effect measure rules out funnel-plot
 1111 and small-study asymmetry tests. We applied no certainty framework such as
 1112 GRADE, since the review produces no pooled effect estimates for such a frame-
 1113 work to rate. Prevalence counts derive from recorded extraction labels and

understate true practice, and no range we quote should be read as an interval estimate rather than as values from named studies.

6. Conclusion

We reviewed 263 primary studies on the safety, artifacts and fidelity of deep learning-based brain MRI reconstruction. Wherever metrics and readers have been compared, agreement between conventional fidelity scores and radiologist judgement is weak, and at most 18 of the 263 studies record the paired metric–reader evaluation capable of exposing that divergence. Every prevalence figure here is a floor derived from screening-level labels rather than a measurement of practice, and although screening agreement was high, the extraction labels those floors rest on carry no reliability estimate of their own. The weakness is structural rather than incidental: the four mechanisms identified under RQ2 explain why, and the safety question is not one that any current evaluation paradigm answers alone.

The contribution we would emphasise is the account of why the gap persists, and we offer it as a hypothesis these data are consistent with rather than as an established structural fact: the venue grouping it rests on is post-hoc, so although the association between code release and reader assessment is strong and would be conventionally significant on a pre-specified partition, we do not read it as a test (Section 4.5). What does not depend on that classification is the magnitude. Only 4.9% of studies both release code and run readers, so computational verifiability and diagnostic verifiability are, as recorded, held by different studies roughly nineteen times in twenty; and no dataset named in this corpus is curated for acute stroke or intracranial haemorrhage. A reconstruction model can therefore satisfy every metric and every public benchmark in current use while the question of whether it preserves a small acute lesion has never been asked. Closing that gap requires benchmarks that contain the pathology at issue and reporting conventions under which no pixel-wise metric appears alone, and until both exist, claims that accelerated reconstruction is diagnostically safe will

1143 continue to rest on measurements that cannot see the failures that matter.
1144 One change is available immediately and needs no new data. Equation (2)
1145 converts any lesion volume into the PSNR movement its erasure would cause, so
1146 an author reporting a reconstruction at a stated acceleration can state, alongside
1147 the global score, the smallest focal change that score is capable of registering
1148 at their image size. For a 100 mm³ infarct that figure is around 0.03 dB, which
1149 is below the precision at which PSNR is conventionally reported. Publishing
1150 that bound next to the metric costs one line, requires no reader and no new
1151 acquisition, and makes explicit what the metric is silent about. It is the cheap-
1152 est useful step towards evaluation that could detect the failure this review is
1153 concerned with.

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1161 reconstruction software contributed financial or non-financial support to this
1162 work.

1163 **Ethics statement**

1164 This review synthesises secondary data from published, de-identified study
1165 reports. It involved no human participants, no animal experimentation and
1166 no collection of private health data, so institutional review board approval and
1167 patient informed consent were not required.

1168 **Data and code availability**

1169 The review protocol, executed search strategy, screening criteria, every
1170 title/abstract and full-text decision with its recorded reason, the com-
1171 pleted PRISMA 2020 checklist, the per-study extraction and QUADAS-
1172 2 worksheets for all 263 included studies, the analysis code, and the
1173 duplicate-screening sample and per-reviewer appraisal worksheets with their
1174 agreement computations (`kappa_worksheet_214_HUMAN_*.csv`, `kappa_hu-`
1175 `man_results.csv`, `reviewer_worksheet_R*.csv` and `quadas2_three_re-`
1176 `viewer_worksheet.csv`) are openly available on the Open Science Frame-
1177 work at <https://doi.org/10.17605/OSF.IO/832Z6> and mirrored at <https://github.com/dmai287/brain-mri-recon-safety-slr>, archived at Zenodo
1178 under DOI 10.5281/zenodo.22100024 (release v1.0.3, commit 8c856ba;
1179 concept DOI 10.5281/zenodo.21836966). Every count reported here is regen-
1180 erable from `included_characteristics.csv` in that release, and the deposit
1181 erable from `included_characteristics.csv` in that release, and the deposit

1182 encodes the same reader/algorithmic partition (57/206) that this article re-
1183 ports, with the four reclassified studies flagged individually. The two figures
1184 generated from that file and the Δ PSNR identity figure are produced by
1185 `5_Code/make_figures.py` and `5_Code/make_dpsnr_figure.py`. The deposit
1186 is retrospective and is documented as such. Full-text PDFs of the included
1187 studies are not redistributed, as publisher licence terms do not permit it; the
1188 DOI list in the released bibliography (`included_studies_263.bib`) identifies
1189 the corpus in full.

1190 **CRediT authorship contribution statement**

1191 **Dat Tat Mai:** Conceptualization, Methodology, Software, Investigation,
1192 Data curation, Formal analysis, Visualization, Project administration, Writing
1193 – original draft, Writing – review & editing. **Thai Viet Pham:** Investiga-
1194 tion, Data curation, Validation, Resources, Writing – review & editing. **Thu**
1195 **Nguyen Thi Dang:** Investigation, Data curation, Validation, Writing – re-
1196 view & editing. **James Jin Kang:** Supervision, Validation, Writing – review
1197 & editing.

1198 **Declaration of competing interest**

1199 The authors declare that they have no known competing financial interests or
1200 personal relationships that could have appeared to influence the work reported
1201 in this paper.

1202 **Declaration of generative AI and AI-assisted technologies in the** 1203 **manuscript preparation process**

1204 During the preparation of this work the authors used generative AI technolo-
1205 gies solely to assist with language editing and proofreading of the manuscript
1206 text. After using these tools, the authors reviewed and edited the content as
1207 needed and take full responsibility for the content of the published article. All

study screening, data extraction, and quality appraisals reported in this review were conducted manually by the authors.

Appendix A. The included studies and how to trace a claim to them

All 263 included primary studies appear in the reference list, so the evidence base is part of the article of record rather than of the supplement alone. Because the synthesis is narrative and most claims are grouped rather than attached to a single study, the reference list alone does not show which studies produced a given number. That mapping is carried by `included_characteristics.csv`, which holds one row per included study with its extracted method family, evaluation paradigm, failure modes, sequences, field strength, reference standard and code availability, each with the verbatim quotation supporting it, so any count, share or crosstab in Section 4 can be reproduced and the contributing studies listed by filtering that file. Supplementary Tables S1–S3 give the same 263 studies as a numbered index by publication year, and `included_characteristics_supplement.csv` adds the per-study QUADAS-2 and reproducibility ratings.

The reference list therefore contains two kinds of entry, and a reader needs a rule to tell them apart. It prints 299: the 263 included studies, carried into the list under PRISMA 2020 item 17, and 36 narrative-only works cited in argument. **An entry appearing in the numbered index of Supplementary Tables S1–S3 is a member of the reviewed corpus; an entry that does not is a narrative citation.** Equivalently, and checkable without the supplement, every narrative citation is cited at a specific point in the running text of Sections 1–6, whereas corpus entries not discussed by name appear without an in-text citation. The two lists were generated by independent mechanisms and agree exactly as sets. We would be glad to move the corpus to a separately labelled supplementary reference list if the editor prefers that production route.

1235 **Appendix B. Supplementary material**

1236 Supplementary material associated with this article (Supplementary Ta-
1237 bles S1–S12 and Supplementary Figure S1) is provided as a separate document.
1238 Tables S1–S3 list all 263 included primary studies with full bibliographic details;
1239 Tables S4–S9 give the comparison against related secondary studies, the search
1240 vocabulary and its derivation, the extraction form, the thematic evidence ar-
1241 chitecture, the four metric-failure mechanisms with their evidence sources, and
1242 the provenance of every released data file. Tables S10–S12 carry the support-
1243 ing detail for Section 4.5: the annotation-layer derivation for fastMRI+ (S10),
1244 the disjoint-strata correction for the sequence contrast (S11), and the venue
1245 cross-tabulation with evaluation-paradigm multiplicity by era (S12).

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